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# CONDITIONAL PROBABILITY AND INDEPENDENCE

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## 1 Introduction

In this chapter, we introduce one of the most important concepts in probability theory, that of conditional probability. The importance of this concept is twofold. In the first place, we are often interested in calculating probabilities when some partial information concerning the result of an experiment is available; in such a situation, the desired probabilities are conditional. Second, even when no partial information is available, conditional probabilities can often be used to compute the desired probabilities more easily.

## 2 Conditional Probabilities

Suppose that we toss 2 dice, and suppose that each of the 36 possible outcomes is equally likely to occur and hence has probability  $\frac{1}{36}$ . Suppose further that we observe that the first die is a 3. Then, given this information, what is the probability that the sum of the 2 dice equals 8? To calculate this probability, we reason as follows: Given that the initial die is a 3, there can be at most 6 possible outcomes of our experiment, namely, (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), and (3, 6). Since each of these outcomes originally had the same probability of occurring, the outcomes should still have equal probabilities. That is, given that the first die is a 3, the (conditional) probability of each of the outcomes (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), and (3, 6) is  $\frac{1}{6}$ , whereas the (conditional) probability of the other 30 points in the sample space is 0. Hence, the desired probability will be  $\frac{1}{6}$ .

If we let  $E$  and  $F$  denote, respectively, the event that the sum of the dice is 8 and the event that the first die is a 3, then the probability just obtained is called the *conditional probability that  $E$  occurs given that  $F$  has occurred* and is denoted by

$$P(E|F)$$

A general formula for  $P(E|F)$  that is valid for all events  $E$  and  $F$  is derived in the same manner: If the event  $F$  occurs, then, in order for  $E$  to occur, it is necessary

that the actual occurrence be a point both in  $E$  and in  $F$ ; that is, it must be in  $EF$ . Now, since we know that  $F$  has occurred, it follows that  $F$  becomes our new, or reduced, sample space; hence, the probability that the event  $EF$  occurs will equal the probability of  $EF$  relative to the probability of  $F$ . That is, we have the following definition.

**Definition**

If  $P(F) > 0$ , then

$$P(E|F) = \frac{P(EF)}{P(F)} \quad (2.1)$$

**Example  
2a**

Joe is 80 percent certain that his missing key is in one of the two pockets of his hanging jacket, being 40 percent certain it is in the left-hand pocket and 40 percent certain it is in the right-hand pocket. If a search of the left-hand pocket does not find the key, what is the conditional probability that it is in the other pocket?

**Solution** If we let  $L$  be the event that the key is in the left-hand pocket of the jacket, and  $R$  be the event that it is in the right-hand pocket, then the desired probability  $P(R|L^c)$  can be obtained as follows:

$$\begin{aligned} P(R|L^c) &= \frac{P(RL^c)}{P(L^c)} \\ &= \frac{P(R)}{1 - P(L)} \\ &= 2/3 \end{aligned}$$

■

If each outcome of a finite sample space  $S$  is equally likely, then, conditional on the event that the outcome lies in a subset  $F \subset S$ , all outcomes in  $F$  become equally likely. In such cases, it is often convenient to compute conditional probabilities of the form  $P(E|F)$  by using  $F$  as the sample space. Indeed, working with this reduced sample space often results in an easier and better understood solution. Our next two examples illustrate this point.

**Example  
2b**

A coin is flipped twice. Assuming that all four points in the sample space  $S = \{(h, h), (h, t), (t, h), (t, t)\}$  are equally likely, what is the conditional probability that both flips land on heads, given that (a) the first flip lands on heads? (b) at least one flip lands on heads?

**Solution** Let  $B = \{(h, h)\}$  be the event that both flips land on heads; let  $F = \{(h, h), (h, t)\}$  be the event that the first flip lands on heads; and let  $A = \{(h, h), (h, t), (t, h)\}$  be the event that at least one flip lands on heads. The probability for (a) can be obtained from

$$\begin{aligned} P(B|F) &= \frac{P(BF)}{P(F)} \\ &= \frac{P(\{(h, h)\})}{P(\{(h, h), (h, t)\})} \\ &= \frac{1/4}{2/4} = 1/2 \end{aligned}$$

For (b), we have

$$\begin{aligned} P(B|A) &= \frac{P(BA)}{P(A)} \\ &= \frac{P(\{(h, h)\})}{P(\{(h, h), (h, t), (t, h)\})} \\ &= \frac{1/4}{3/4} = 1/3 \end{aligned}$$

Thus, the conditional probability that both flips land on heads given that the first one does is  $1/2$ , whereas the conditional probability that both flips land on heads given that at least one does is only  $1/3$ . Many students initially find this latter result surprising. They reason that given that at least one flip lands on heads, there are two possible results: Either they both land on heads or only one does. Their mistake, however, is in assuming that these two possibilities are equally likely. Initially there are 4 equally likely outcomes. Because the information that at least one flip lands on heads is equivalent to the information that the outcome is not  $(t, t)$ , we are left with the 3 equally likely outcomes  $(h, h)$ ,  $(h, t)$ ,  $(t, h)$ , only one of which results in both flips landing on heads. ■

**Example  
2c**

In the card game bridge, the 52 cards are dealt out equally to 4 players—called East, West, North, and South. If North and South have a total of 8 spades among them, what is the probability that East has 3 of the remaining 5 spades?

**Solution** Probably the easiest way to compute the desired probability is to work with the reduced sample space. That is, given that North–South have a total of 8 spades among their 26 cards, there remains a total of 26 cards, exactly 5 of them being spades, to be distributed among the East–West hands. Since each distribution is equally likely, it follows that the conditional probability that East will have exactly 3 spades among his or her 13 cards is

$$\frac{\binom{5}{3} \binom{21}{10}}{\binom{26}{13}} \approx .339 \quad \blacksquare$$

Multiplying both sides of Equation (2.1) by  $P(F)$ , we obtain

$$P(EF) = P(F)P(E|F) \quad (2.2)$$

In words, Equation (2.2) states that the probability that both  $E$  and  $F$  occur is equal to the probability that  $F$  occurs multiplied by the conditional probability of  $E$  given that  $F$  occurred. Equation (2.2) is often quite useful in computing the probability of the intersection of events.

**Example  
2d**

Celine is undecided as to whether to take a French course or a chemistry course. She estimates that her probability of receiving an A grade would be  $\frac{1}{2}$  in a French course and  $\frac{2}{3}$  in a chemistry course. If Celine decides to base her decision on the flip of a fair coin, what is the probability that she gets an A in chemistry?

**Solution** Let  $C$  be the event that Celine takes chemistry and  $A$  denote the event that she receives an A in whatever course she takes, then the desired probability is  $P(CA)$ , which is calculated by using Equation (2.2) as follows:

$$\begin{aligned} P(CA) &= P(C)P(A|C) \\ &= \left(\frac{1}{2}\right)\left(\frac{2}{3}\right) = \frac{1}{3} \end{aligned}$$

**Example 2e**

Suppose that an urn contains 8 red balls and 4 white balls. We draw 2 balls from the urn without replacement. (a) If we assume that at each draw, each ball in the urn is equally likely to be chosen, what is the probability that both balls drawn are red? (b) Now suppose that the balls have different weights, with each red ball having weight  $r$  and each white ball having weight  $w$ . Suppose that the probability that a given ball in the urn is the next one selected is its weight divided by the sum of the weights of all balls currently in the urn. Now what is the probability that both balls are red?

**Solution** Let  $R_1$  and  $R_2$  denote, respectively, the events that the first and second balls drawn are red. Now, given that the first ball selected is red, there are 7 remaining red balls and 4 white balls, so  $P(R_2|R_1) = \frac{7}{11}$ . As  $P(R_1)$  is clearly  $\frac{8}{12}$ , the desired probability is

$$\begin{aligned} P(R_1R_2) &= P(R_1)P(R_2|R_1) \\ &= \left(\frac{2}{3}\right)\left(\frac{7}{11}\right) = \frac{14}{33} \end{aligned}$$

Of course, this probability could have been computed by  $P(R_1R_2) = \frac{\binom{8}{2}}{\binom{12}{2}}$ .

For part (b), we again let  $R_i$  be the event that the  $i$ th ball chosen is red and use

$$P(R_1R_2) = P(R_1)P(R_2|R_1)$$

Now, number the red balls, and let  $B_i$ ,  $i = 1, \dots, 8$  be the event that the first ball drawn is red ball number  $i$ . Then

$$P(R_1) = P(\cup_{i=1}^8 B_i) = \sum_{i=1}^8 P(B_i) = 8 \frac{r}{8r + 4w}$$

Moreover, given that the first ball is red, the urn then contains 7 red and 4 white balls. Thus, by an argument similar to the preceding one,

$$P(R_2|R_1) = \frac{7r}{7r + 4w}$$

Hence, the probability that both balls are red is

$$P(R_1R_2) = \frac{8r}{8r + 4w} \frac{7r}{7r + 4w}$$

A generalization of Equation (2.2), which provides an expression for the probability of the intersection of an arbitrary number of events, is sometimes referred to as the *multiplication rule*.

**The multiplication rule**

$$P(E_1E_2E_3 \cdots E_n) = P(E_1)P(E_2|E_1)P(E_3|E_1E_2) \cdots P(E_n|E_1 \cdots E_{n-1})$$

To prove the multiplication rule, just apply the definition of conditional probability to its right-hand side, giving

$$P(E_1) \frac{P(E_1 E_2)}{P(E_1)} \frac{P(E_1 E_2 E_3)}{P(E_1 E_2)} \cdots \frac{P(E_1 E_2 \cdots E_n)}{P(E_1 E_2 \cdots E_{n-1})} = P(E_1 E_2 \cdots E_n)$$

**Example  
2f**

For example, consider that  $P_N$ , the probability that there are no matches when  $N$  people randomly select from among their own  $N$  hats, is given by

$$P_N = \sum_{i=0}^N (-1)^i / i!$$

What is the probability that exactly  $k$  of the  $N$  people have matches?

**Solution** Let us fix our attention on a particular set of  $k$  people and determine the probability that these  $k$  individuals have matches and no one else does. Letting  $E$  denote the event that everyone in this set has a match, and letting  $G$  be the event that none of the other  $N - k$  people have a match, we have

$$P(EG) = P(E)P(G|E)$$

Now, let  $F_i$ ,  $i = 1, \dots, k$ , be the event that the  $i$ th member of the set has a match. Then

$$\begin{aligned} P(E) &= P(F_1 F_2 \cdots F_k) \\ &= P(F_1)P(F_2|F_1)P(F_3|F_1 F_2) \cdots P(F_k|F_1 \cdots F_{k-1}) \\ &= \frac{1}{N} \frac{1}{N-1} \frac{1}{N-2} \cdots \frac{1}{N-k+1} \\ &= \frac{(N-k)!}{N!} \end{aligned}$$

Given that everyone in the set of  $k$  has a match, the other  $N - k$  people will be randomly choosing among their own  $N - k$  hats, so the probability that none of them has a match is equal to the probability of no matches in a problem having  $N - k$  people choosing among their own  $N - k$  hats. Therefore,

$$P(G|E) = P_{N-k} = \sum_{i=0}^{N-k} (-1)^i / i!$$

showing that the probability that a specified set of  $k$  people have matches and no one else does is

$$P(EG) = \frac{(N-k)!}{N!} P_{N-k}$$

Because there will be exactly  $k$  matches if the preceding is true for any of the  $\binom{N}{k}$  sets of  $k$  individuals, the desired probability is

$$\begin{aligned} P(\text{exactly } k \text{ matches}) &= P_{N-k} / k! \\ &\approx e^{-1} / k! \quad \text{when } N \text{ is large} \end{aligned} \quad \blacksquare$$

**Example  
2g**

An ordinary deck of 52 playing cards is randomly divided into 4 piles of 13 cards each. Compute the probability that each pile has exactly 1 ace.

**Solution** Define events  $E_i, i = 1, 2, 3, 4$ , as follows:

$E_1 = \{\text{the ace of spades is in any one of the piles}\}$

$E_2 = \{\text{the ace of spades and the ace of hearts are in different piles}\}$

$E_3 = \{\text{the aces of spades, hearts, and diamonds are all in different piles}\}$

$E_4 = \{\text{all 4 aces are in different piles}\}$

The desired probability is  $P(E_1E_2E_3E_4)$ , and by the multiplication rule,

$$P(E_1E_2E_3E_4) = P(E_1)P(E_2|E_1)P(E_3|E_1E_2)P(E_4|E_1E_2E_3)$$

Now,

$$P(E_1) = 1$$

since  $E_1$  is the sample space  $S$ . To determine  $P(E_2|E_1)$ , consider the pile that contains the ace of spades. Because its remaining 12 cards are equally likely to be any 12 of the remaining 51 cards, the probability that the ace of hearts is among them is  $12/51$ , giving that

$$P(E_2|E_1) = 1 - \frac{12}{51} = \frac{39}{51}$$

Also, given that the ace of spades and ace of hearts are in different piles, it follows that the set of the remaining 24 cards of these two piles is equally likely to be any set of 24 of the remaining 50 cards. As the probability that the ace of diamonds is one of these 24 is  $24/50$ , we see that

$$P(E_3|E_1E_2) = 1 - \frac{24}{50} = \frac{26}{50}$$

Because the same logic as used in the preceding yields that

$$P(E_4|E_1E_2E_3) = 1 - \frac{36}{49} = \frac{13}{49}$$

the probability that each pile has exactly 1 ace is

$$P(E_1E_2E_3E_4) = \frac{39 \cdot 26 \cdot 13}{51 \cdot 50 \cdot 49} \approx .105$$

That is, there is approximately a 10.5 percent chance that each pile will contain an ace. (Problem 13 gives another way of using the multiplication rule to solve this problem.) ■

**Remarks** Our definition of  $P(E|F)$  is consistent with the interpretation of probability as being a long-run relative frequency. To see this, suppose that  $n$  repetitions of the experiment are to be performed, where  $n$  is large. We claim that if we consider only those experiments in which  $F$  occurs, then  $P(E|F)$  will equal the long-run proportion of them in which  $E$  also occurs. To verify this statement, note that since  $P(F)$  is the long-run proportion of experiments in which  $F$  occurs, it follows that in the  $n$  repetitions of the experiment,  $F$  will occur approximately  $nP(F)$  times. Similarly, in approximately  $nP(EF)$  of these experiments, both  $E$  and  $F$  will occur. Hence, out of the approximately  $nP(F)$  experiments in which  $F$  occurs, the proportion of them in which  $E$  also occurs is approximately equal to

$$\frac{nP(EF)}{nP(F)} = \frac{P(EF)}{P(F)}$$

Because this approximation becomes exact as  $n$  becomes larger and larger, we have the appropriate definition of  $P(E|F)$ .

### 3 Bayes's Formula

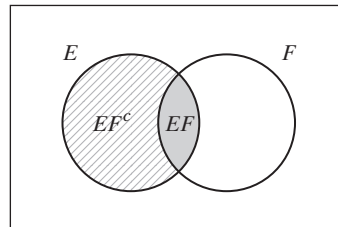
Let  $E$  and  $F$  be events. We may express  $E$  as

$$E = EF \cup EF^c$$

for, in order for an outcome to be in  $E$ , it must either be in both  $E$  and  $F$  or be in  $E$  but not in  $F$ . (See Figure 1.) As  $EF$  and  $EF^c$  are clearly mutually exclusive, we have, by Axiom 3,

$$\begin{aligned} P(E) &= P(EF) + P(EF^c) \\ &= P(E|F)P(F) + P(E|F^c)P(F^c) \\ &= P(E|F)P(F) + P(E|F^c)[1 - P(F)] \end{aligned} \quad (3.1)$$

Equation (3.1) states that the probability of the event  $E$  is a weighted average of the conditional probability of  $E$  given that  $F$  has occurred and the conditional probability of  $E$  given that  $F$  has not occurred—each conditional probability being given as much weight as the event on which it is conditioned has of occurring. This is an extremely useful formula, because its use often enables us to determine the probability of an event by first “conditioning” upon whether or not some second event has occurred. That is, there are many instances in which it is difficult to compute the probability of an event directly, but it is straightforward to compute it once we know whether or not some second event has occurred. We illustrate this idea with some examples.



**Figure 1**  $E = EF \cup EF^c$ .  $EF$  = Shaded Area;  $EF^c$  = Striped Area.

#### Example 3a

##### (Part 1)

An insurance company believes that people can be divided into two classes: those who are accident prone and those who are not. The company's statistics show that an accident-prone person will have an accident at some time within a fixed 1-year period with probability .4, whereas this probability decreases to .2 for a person who is not accident prone. If we assume that 30 percent of the population is accident prone, what is the probability that a new policyholder will have an accident within a year of purchasing a policy?

**Solution** We shall obtain the desired probability by first conditioning upon whether or not the policyholder is accident prone. Let  $A_1$  denote the event that the policyholder will have an accident within a year of purchasing the policy, and let  $A$  denote

the event that the policyholder is accident prone. Hence, the desired probability is given by

$$\begin{aligned} P(A_1) &= P(A_1|A)P(A) + P(A_1|A^c)P(A^c) \\ &= (.4)(.3) + (.2)(.7) = .26 \end{aligned} \quad \blacksquare$$

**Example 3a**

**(Part 2)**

Suppose that a new policyholder has an accident within a year of purchasing a policy. What is the probability that he or she is accident prone?

**Solution** The desired probability is

$$\begin{aligned} P(A|A_1) &= \frac{P(AA_1)}{P(A_1)} \\ &= \frac{P(A)P(A_1|A)}{P(A_1)} \\ &= \frac{(.3)(.4)}{.26} = \frac{6}{13} \end{aligned} \quad \blacksquare$$

**Example 3b**

Consider the following game played with an ordinary deck of 52 playing cards: The cards are shuffled and then turned over one at a time. At any time, the player can guess that the next card to be turned over will be the ace of spades; if it is, then the player wins. In addition, the player is said to win if the ace of spades has not yet appeared when only one card remains and no guess has yet been made. What is a good strategy? What is a bad strategy?

**Solution** Every strategy has probability  $1/52$  of winning! To show this, we will use induction to prove the stronger result that for an  $n$  card deck, one of whose cards is the ace of spades, the probability of winning is  $1/n$ , no matter what strategy is employed. Since this is clearly true for  $n = 1$ , assume it to be true for an  $n - 1$  card deck, and now consider an  $n$  card deck. Fix any strategy, and let  $p$  denote the probability that the strategy guesses that the first card is the ace of spades. Given that it does, the player's probability of winning is  $1/n$ . If, however, the strategy does not guess that the first card is the ace of spades, then the probability that the player wins is the probability that the first card is not the ace of spades, namely,  $(n - 1)/n$ , multiplied by the conditional probability of winning given that the first card is not the ace of spades. But this latter conditional probability is equal to the probability of winning when using an  $n - 1$  card deck containing a single ace of spades; it is thus, by the induction hypothesis,  $1/(n - 1)$ . Hence, given that the strategy does not guess the first card, the probability of winning is

$$\frac{n - 1}{n} \cdot \frac{1}{n - 1} = \frac{1}{n}$$

Thus, letting  $G$  be the event that the first card is guessed, we obtain

$$\begin{aligned} P\{\text{win}\} &= P\{\text{win}|G\}P(G) + P\{\text{win}|G^c\}(1 - P(G)) = \frac{1}{n}p + \frac{1}{n}(1 - p) \\ &= \frac{1}{n} \end{aligned} \quad \blacksquare$$

**Example 3c**

In answering a question on a multiple-choice test, a student either knows the answer or guesses. Let  $p$  be the probability that the student knows the answer and  $1 - p$  be the probability that the student guesses. Assume that a student who guesses at

the answer will be correct with probability  $1/m$ , where  $m$  is the number of multiple-choice alternatives. What is the conditional probability that a student knew the answer to a question given that he or she answered it correctly?

**Solution** Let  $C$  and  $K$  denote, respectively, the events that the student answers the question correctly and the event that he or she actually knows the answer. Now,

$$\begin{aligned} P(K|C) &= \frac{P(KC)}{P(C)} \\ &= \frac{P(C|K)P(K)}{P(C|K)P(K) + P(C|K^c)P(K^c)} \\ &= \frac{p}{p + (1/m)(1 - p)} \\ &= \frac{mp}{1 + (m - 1)p} \end{aligned}$$

For example, if  $m = 5, p = \frac{1}{2}$ , then the probability that the student knew the answer to a question he or she answered correctly is  $\frac{5}{6}$ . ■

**Example  
3d**

A laboratory blood test is 95 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a “false positive” result for 1 percent of the healthy persons tested. (That is, if a healthy person is tested, then, with probability .01, the test result will imply that he or she has the disease.) If .5 percent of the population actually has the disease, what is the probability that a person has the disease given that the test result is positive?

**Solution** Let  $D$  be the event that the person tested has the disease and  $E$  the event that the test result is positive. Then the desired probability is

$$\begin{aligned} P(D|E) &= \frac{P(DE)}{P(E)} \\ &= \frac{P(E|D)P(D)}{P(E|D)P(D) + P(E|D^c)P(D^c)} \\ &= \frac{(.95)(.005)}{(.95)(.005) + (.01)(.995)} \\ &= \frac{95}{294} \approx .323 \end{aligned}$$

Thus, only 32 percent of those persons whose test results are positive actually have the disease. Many students are often surprised at this result (they expect the percentage to be much higher, since the blood test seems to be a good one), so it is probably worthwhile to present a second argument that, although less rigorous than the preceding one, is probably more revealing. We now do so.

Since .5 percent of the population actually has the disease, it follows that, on the average, 1 person out of every 200 tested will have it. The test will correctly confirm that this person has the disease with probability .95. Thus, on the average, out of every 200 persons tested, the test will correctly confirm that .95 person has the disease. On the other hand, out of the (on the average) 199 healthy people, the test will incorrectly state that (199)(.01) of these people have the disease. Hence, for every .95 diseased person that the test correctly states is ill, there are (on the average) (199)(.01) healthy persons who the test incorrectly states are ill. Thus,

the proportion of time that the test result is correct when it states that a person is ill is

$$\frac{.95}{.95 + (199)(.01)} = \frac{95}{294} \approx .323 \quad \blacksquare$$

Equation (3.1) is also useful when one has to reassess one's personal probabilities in the light of additional information. For instance, consider the examples that follow.

**Example**  
**3e**

Consider a medical practitioner pondering the following dilemma: "If I'm at least 80 percent certain that my patient has this disease, then I always recommend surgery, whereas if I'm not quite as certain, then I recommend additional tests that are expensive and sometimes painful. Now, initially I was only 60 percent certain that Jones had the disease, so I ordered the series A test, which always gives a positive result when the patient has the disease and almost never does when he is healthy. The test result was positive, and I was all set to recommend surgery when Jones informed me, for the first time, that he was diabetic. This information complicates matters because, although it doesn't change my original 60 percent estimate of his chances of having the disease in question, it does affect the interpretation of the results of the A test. This is so because the A test, while never yielding a positive result when the patient is healthy, does unfortunately yield a positive result 30 percent of the time in the case of *diabetic* patients who are not suffering from the disease. Now what do I do? More tests or immediate surgery?"

**Solution** In order to decide whether or not to recommend surgery, the doctor should first compute her updated probability that Jones has the disease given that the A test result was positive. Let  $D$  denote the event that Jones has the disease and  $E$  the event that the A test result is positive. The desired conditional probability is then

$$\begin{aligned} P(D|E) &= \frac{P(DE)}{P(E)} \\ &= \frac{P(D)P(E|D)}{P(E|D)P(D) + P(E|D^c)P(D^c)} \\ &= \frac{(.6)1}{1(.6) + (.3)(.4)} \\ &= .833 \end{aligned}$$

Note that we have computed the probability of a positive test result by conditioning on whether or not Jones has the disease and then using the fact that because Jones is a diabetic, his conditional probability of a positive result given that he does not have the disease,  $P(E|D^c)$ , equals .3. Hence, as the doctor should now be more than 80 percent certain that Jones has the disease, she should recommend surgery. ■

**Example**  
**3f**

At a certain stage of a criminal investigation, the inspector in charge is 60 percent convinced of the guilt of a certain suspect. Suppose, however, that a *new* piece of evidence which shows that the criminal has a certain characteristic (such as left-handedness, baldness, or brown hair) is uncovered. If 20 percent of the population possesses this characteristic, how certain of the guilt of the suspect should the inspector now be if it turns out that the suspect has the characteristic?

**Solution** Letting  $G$  denote the event that the suspect is guilty and  $C$  the event that he possesses the characteristic of the criminal, we have

$$\begin{aligned} P(G|C) &= \frac{P(GC)}{P(C)} \\ &= \frac{P(C|G)P(G)}{P(C|G)P(G) + P(C|G^c)P(G^c)} \\ &= \frac{1(.6)}{1(.6) + (.2)(.4)} \\ &\approx .882 \end{aligned}$$

where we have supposed that the probability of the suspect having the characteristic if he is, in fact, innocent is equal to .2, the proportion of the population possessing the characteristic. ■

**Example 3g**

In the world bridge championships held in Buenos Aires in May 1965, the famous British bridge partnership of Terrence Reese and Boris Schapiro was accused of cheating by using a system of finger signals that could indicate the number of hearts held by the players. Reese and Schapiro denied the accusation, and eventually a hearing was held by the British bridge league. The hearing was in the form of a legal proceeding with prosecution and defense teams, both having the power to call and cross-examine witnesses. During the course of the proceeding, the prosecutor examined specific hands played by Reese and Schapiro and claimed that their playing these hands was consistent with the hypothesis that they were guilty of having illicit knowledge of the heart suit. At this point, the defense attorney pointed out that their play of these hands was also perfectly consistent with their standard line of play. However, the prosecution then argued that as long as their play was consistent with the hypothesis of guilt, it must be counted as evidence toward that hypothesis. What do you think of the reasoning of the prosecution?

**Solution** The problem is basically one of determining how the introduction of new evidence (in this example, the playing of the hands) affects the probability of a particular hypothesis. If we let  $H$  denote a particular hypothesis (such as the hypothesis that Reese and Schapiro are guilty) and  $E$  the new evidence, then

$$\begin{aligned} P(H|E) &= \frac{P(HE)}{P(E)} \\ &= \frac{P(E|H)P(H)}{P(E|H)P(H) + P(E|H^c)[1 - P(H)]} \end{aligned} \quad (3.2)$$

where  $P(H)$  is our evaluation of the likelihood of the hypothesis before the introduction of the new evidence. The new evidence will be in support of the hypothesis whenever it makes the hypothesis more likely—that is, whenever  $P(H|E) \geq P(H)$ . From Equation (3.2), this will be the case whenever

$$P(E|H) \geq P(E|H)P(H) + P(E|H^c)[1 - P(H)]$$

or, equivalently, whenever

$$P(E|H) \geq P(E|H^c)$$

In other words, any new evidence can be considered to be in support of a particular hypothesis only if its occurrence is more likely when the hypothesis is true than when it is false. In fact, the new probability of the hypothesis depends on its initial probability and the ratio of these conditional probabilities, since, from Equation (3.2),

$$P(H|E) = \frac{P(H)}{P(H) + [1 - P(H)] \frac{P(E|H^c)}{P(E|H)}}$$

Hence, in the problem under consideration, the play of the cards can be considered to support the hypothesis of guilt only if such play would have been more likely if the partnership were cheating than if it were not. As the prosecutor never made this claim, his assertion that the evidence is in support of the guilt hypothesis is invalid. ■

**Example  
3h**

Twins can be either identical or fraternal. Identical, also called monozygotic, twins form when a single fertilized egg splits into two genetically identical parts. Consequently, identical twins always have the same set of genes. Fraternal, also called dizygotic, twins develop when two eggs are fertilized and implant in the uterus. The genetic connection of fraternal twins is no more or less the same as siblings born at separate times. A Los Angeles County, California, scientist wishing to know the current fraction of twin pairs born in the county that are identical twins has assigned a county statistician to study this issue. The statistician initially requested each hospital in the county to record all twin births, indicating whether or not the resulting twins were identical. The hospitals, however, told her that to determine whether newborn twins were identical was not a simple task, as it involved the permission of the twins' parents to perform complicated and expensive DNA studies that the hospitals could not afford. After some deliberation, the statistician just asked the hospitals for data listing all twin births along with an indication as to whether the twins were of the same sex. When such data indicated that approximately 64 percent of twin births were same-sexed, the statistician declared that approximately 28 percent of all twins were identical. How did she come to this conclusion?

**Solution** The statistician reasoned that identical twins are always of the same sex, whereas fraternal twins, having the same relationship to each other as any pair of siblings, will have probability  $1/2$  of being of the same sex. Letting  $I$  be the event that a pair of twins is identical, and  $SS$  be the event that a pair of twins is of the same sex, she computed the probability  $P(SS)$  by conditioning on whether the twin pair was identical. This gave

$$P(SS) = P(SS|I)P(I) + P(SS|I^c)P(I^c)$$

or

$$P(SS) = 1 \times P(I) + \frac{1}{2} \times [1 - P(I)] = \frac{1}{2} + \frac{1}{2} P(I)$$

which, using that  $P(SS) \approx .64$  yielded the result

$$P(I) \approx .28$$

The change in the probability of a hypothesis when new evidence is introduced can be expressed compactly in terms of the change in the *odds* of that hypothesis, where the concept of odds is defined as follows.

**Definition**

The odds of an event  $A$  are defined by

$$\frac{P(A)}{P(A^c)} = \frac{P(A)}{1 - P(A)}$$

That is, the odds of an event  $A$  tell how much more likely it is that the event  $A$  occurs than it is that it does not occur. For instance, if  $P(A) = \frac{2}{3}$ , then  $P(A) = 2P(A^c)$ , so the odds are 2. If the odds are equal to  $\alpha$ , then it is common to say that the odds are “ $\alpha$  to 1” in favor of the hypothesis.

Consider now a hypothesis  $H$  that is true with probability  $P(H)$ , and suppose that new evidence  $E$  is introduced. Then, the conditional probabilities, given the evidence  $E$ , that  $H$  is true and that  $H$  is not true are respectively given by

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)} \quad P(H^c|E) = \frac{P(E|H^c)P(H^c)}{P(E)}$$

Therefore, the new odds after the evidence  $E$  has been introduced are

$$\frac{P(H|E)}{P(H^c|E)} = \frac{P(H)}{P(H^c)} \frac{P(E|H)}{P(E|H^c)} \quad (3.3)$$

That is, the new value of the odds of  $H$  is the old value multiplied by the ratio of the conditional probability of the new evidence given that  $H$  is true to the conditional probability given that  $H$  is not true. Thus, Equation (3.3) verifies the result of Example 3f, since the odds, and thus the probability of  $H$ , increase whenever the new evidence is more likely when  $H$  is true than when it is false. Similarly, the odds decrease whenever the new evidence is more likely when  $H$  is false than when it is true.

**Example 3i**

An urn contains two type  $A$  coins and one type  $B$  coin. When a type  $A$  coin is flipped, it comes up heads with probability  $1/4$ , whereas when a type  $B$  coin is flipped, it comes up heads with probability  $3/4$ . A coin is randomly chosen from the urn and flipped. Given that the flip landed on heads, what is the probability that it was a type  $A$  coin?

**Solution** Let  $A$  be the event that a type  $A$  coin was flipped, and let  $B = A^c$  be the event that a type  $B$  coin was flipped. We want  $P(A|\text{heads})$ , where heads is the event that the flip landed on heads. From Equation (3.3), we see that

$$\begin{aligned} \frac{P(A|\text{heads})}{P(A^c|\text{heads})} &= \frac{P(A)}{P(B)} \frac{P(\text{heads}|A)}{P(\text{heads}|B)} \\ &= \frac{2/3}{1/3} \frac{1/4}{3/4} \\ &= 2/3 \end{aligned}$$

Hence, the odds are  $2/3 : 1$ , or, equivalently, the probability is  $2/5$  that a type  $A$  coin was flipped. ■

Equation (3.1) may be generalized as follows: Suppose that  $F_1, F_2, \dots, F_n$  are mutually exclusive events such that

$$\bigcup_{i=1}^n F_i = S$$

In other words, exactly one of the events  $F_1, F_2, \dots, F_n$  must occur. By writing

$$E = \bigcup_{i=1}^n EF_i$$

and using the fact that the events  $EF_i, i = 1, \dots, n$  are mutually exclusive, we obtain

$$\begin{aligned} P(E) &= \sum_{i=1}^n P(EF_i) \\ &= \sum_{i=1}^n P(E|F_i)P(F_i) \end{aligned} \quad (3.4)$$

Thus, Equation (3.4), often referred to as the *law of total probability*, shows how, for given events  $F_1, F_2, \dots, F_n$ , of which one and only one must occur, we can compute  $P(E)$  by first conditioning on which one of the  $F_i$  occurs. That is, Equation (3.4) states that  $P(E)$  is equal to a weighted average of  $P(E|F_i)$ , each term being weighted by the probability of the event on which it is conditioned.

**Example**  
**3j**

For example, for a randomly shuffled deck, if the card following the first ace is some specified card, then as per a combinatorial argument the probability is  $\frac{1}{52}$ . Here is a probabilistic argument based on conditioning: Let  $E$  be the event that the card following the first ace is some specified card, say, card  $x$ . To compute  $P(E)$ , we ignore card  $x$  and condition on the relative ordering of the other 51 cards in the deck. Letting  $\mathbf{O}$  be the ordering gives

$$P(E) = \sum_{\mathbf{O}} P(E|\mathbf{O})P(\mathbf{O})$$

Now, given  $\mathbf{O}$ , there are 52 possible orderings of the cards, corresponding to having card  $x$  being the  $i$ th card in the deck,  $i = 1, \dots, 52$ . But because all 52! possible orderings were initially equally likely, it follows that, conditional on  $\mathbf{O}$ , each of the 52 remaining possible orderings is equally likely. Because card  $x$  will follow the first ace for only one of these orderings, we have  $P(E|\mathbf{O}) = 1/52$ , implying that  $P(E) = 1/52$ . ■

Again, let  $F_1, \dots, F_n$  be a set of mutually exclusive and exhaustive events (meaning that exactly one of these events must occur).

Suppose now that  $E$  has occurred and we are interested in determining which one of the  $F_j$  also occurred. Then, by Equation (3.4), we have the following proposition.

**Proposition**  
**3.1**

$$\begin{aligned} P(F_j|E) &= \frac{P(EF_j)}{P(E)} \\ &= \frac{P(E|F_j)P(F_j)}{\sum_{i=1}^n P(E|F_i)P(F_i)} \end{aligned} \quad (3.5)$$

Equation (3.5) is known as *Bayes's formula*, after the English philosopher Thomas Bayes. If we think of the events  $F_j$  as being possible “hypotheses” about some subject matter, then Bayes's formula may be interpreted as showing us how opinions about these hypotheses held before the experiment was carried out [that is, the  $P(F_j)$ ] should be modified by the evidence produced by the experiment.

**Example**  
**3k**

A plane is missing, and it is presumed that it was equally likely to have gone down in any of 3 possible regions. Let  $1 - \beta_i$ ,  $i = 1, 2, 3$ , denote the probability that the plane will be found upon a search of the  $i$ th region when the plane is, in fact, in that region. (The constants  $\beta_i$  are called *overlook probabilities*, because they represent the probability of overlooking the plane; they are generally attributable to the geographical and environmental conditions of the regions.) What is the conditional probability that the plane is in the  $i$ th region given that a search of region 1 is unsuccessful?

**Solution** Let  $R_i$ ,  $i = 1, 2, 3$ , be the event that the plane is in region  $i$ , and let  $E$  be the event that a search of region 1 is unsuccessful. From Bayes's formula, we obtain

$$\begin{aligned} P(R_1|E) &= \frac{P(ER_1)}{P(E)} \\ &= \frac{P(E|R_1)P(R_1)}{\sum_{i=1}^3 P(E|R_i)P(R_i)} \\ &= \frac{(\beta_1)^{\frac{1}{3}}}{(\beta_1)^{\frac{1}{3}} + (1)^{\frac{1}{3}} + (1)^{\frac{1}{3}}} \\ &= \frac{\beta_1}{\beta_1 + 2} \end{aligned}$$

For  $j = 2, 3$ ,

$$\begin{aligned} P(R_j|E) &= \frac{P(E|R_j)P(R_j)}{P(E)} \\ &= \frac{(1)^{\frac{1}{3}}}{(\beta_1)^{\frac{1}{3}} + \frac{1}{3} + \frac{1}{3}} \\ &= \frac{1}{\beta_1 + 2} \quad j = 2, 3 \end{aligned}$$

Note that the updated (that is, the conditional) probability that the plane is in region  $j$ , given the information that a search of region 1 did not find it, is greater than the initial probability that it was in region  $j$  when  $j \neq 1$  and is less than the initial probability when  $j = 1$ . This statement is certainly intuitive, since not finding the plane in region 1 would seem to decrease its chance of being in that region and increase its chance of being elsewhere. Further, the conditional probability that the plane is in region 1 given an unsuccessful search of that region is an increasing function of the overlook probability  $\beta_1$ . This statement is also intuitive, since the larger  $\beta_1$  is, the more it is reasonable to attribute the unsuccessful search to “bad luck” as opposed to the plane's not being there. Similarly,  $P(R_j|E)$ ,  $j \neq 1$ , is a decreasing function of  $\beta_1$ . ■

The next example has often been used by unscrupulous probability students to win money from their less enlightened friends.

**Example  
3l**

Suppose that we have 3 cards that are identical in form, except that both sides of the first card are colored red, both sides of the second card are colored black, and one side of the third card is colored red and the other side black. The 3 cards are mixed up in a hat, and 1 card is randomly selected and put down on the ground. If the upper side of the chosen card is colored red, what is the probability that the other side is colored black?

**Solution** Let  $RR$ ,  $BB$ , and  $RB$  denote, respectively, the events that the chosen card is all red, all black, or the red–black card. Also, let  $R$  be the event that the upturned side of the chosen card is red. Then, the desired probability is obtained by

$$\begin{aligned} P(RB|R) &= \frac{P(RB \cap R)}{P(R)} \\ &= \frac{P(R|RB)P(RB)}{P(R|RR)P(RR) + P(R|RB)P(RB) + P(R|BB)P(BB)} \\ &= \frac{\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)}{(1)\left(\frac{1}{3}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{3}\right) + 0\left(\frac{1}{3}\right)} = \frac{1}{3} \end{aligned}$$

Hence, the answer is  $\frac{1}{3}$ . Some students guess  $\frac{1}{2}$  as the answer by incorrectly reasoning that given that a red side appears, there are two equally likely possibilities: that the card is the all-red card or the red–black card. Their mistake, however, is in assuming that these two possibilities are equally likely. For, if we think of each card as consisting of two distinct sides, then we see that there are 6 equally likely outcomes of the experiment—namely,  $R_1, R_2, B_1, B_2, R_3, B_3$ —where the outcome is  $R_1$  if the first side of the all-red card is turned face up,  $R_2$  if the second side of the all-red card is turned face up,  $R_3$  if the red side of the red–black card is turned face up, and so on. Since the other side of the upturned red side will be black only if the outcome is  $R_3$ , we see that the desired probability is the conditional probability of  $R_3$  given that either  $R_1$  or  $R_2$  or  $R_3$  occurred, which obviously equals  $\frac{1}{3}$ . ■

**Example  
3m**

A new couple, known to have two children, has just moved into town. Suppose that the mother is encountered walking with one of her children. If this child is a girl, what is the probability that both children are girls?

**Solution** Let us start by defining the following events:

$G_1$ : the first (that is, the oldest) child is a girl.

$G_2$ : the second child is a girl.

$G$ : the child seen with the mother is a girl.

Also, let  $B_1, B_2$ , and  $B$  denote similar events, except that “girl” is replaced by “boy.” Now, the desired probability is  $P(G_1 G_2 | G)$ , which can be expressed as follows:

$$\begin{aligned} P(G_1 G_2 | G) &= \frac{P(G_1 G_2 G)}{P(G)} \\ &= \frac{P(G_1 G_2)}{P(G)} \end{aligned}$$

Also,

$$\begin{aligned} P(G) &= P(G|G_1G_2)P(G_1G_2) + P(G|G_1B_2)P(G_1B_2) \\ &\quad + P(G|B_1G_2)P(B_1G_2) + P(G|B_1B_2)P(B_1B_2) \\ &= P(G_1G_2) + P(G|G_1B_2)P(G_1B_2) + P(G|B_1G_2)P(B_1G_2) \end{aligned}$$

where the final equation used the results  $P(G|G_1G_2) = 1$  and  $P(G|B_1B_2) = 0$ . If we now make the usual assumption that all 4 gender possibilities are equally likely, then we see that

$$\begin{aligned} P(G_1G_2|G) &= \frac{\frac{1}{4}}{\frac{1}{4} + P(G|G_1B_2)/4 + P(G|B_1G_2)/4} \\ &= \frac{1}{1 + P(G|G_1B_2) + P(G|B_1G_2)} \end{aligned}$$

Thus, the answer depends on whatever assumptions we want to make about the conditional probabilities that the child seen with the mother is a girl given the event  $G_1B_2$  and that the child seen with the mother is a girl given the event  $G_2B_1$ . For instance, if we want to assume, on the one hand, that, independently of the genders of the children, the child walking with the mother is the elder child with some probability  $p$ , then it would follow that

$$P(G|G_1B_2) = p = 1 - P(G|B_1G_2)$$

implying under this scenario that

$$P(G_1G_2|G) = \frac{1}{2}$$

If, on the other hand, we were to assume that if the children are of different genders, then the mother would choose to walk with the girl with probability  $q$ , independently of the birth order of the children, then we would have

$$P(G|G_1B_2) = P(G|B_1G_2) = q$$

implying that

$$P(G_1G_2|G) = \frac{1}{1 + 2q}$$

For instance, if we took  $q = 1$ , meaning that the mother would always choose to walk with a daughter, then the conditional probability that she has two daughters would be  $\frac{1}{3}$ , which is in accord with Example 2b because seeing the mother with a daughter is now equivalent to the event that she has at least one daughter.

Hence, as stated, the problem is incapable of solution. Indeed, even when the usual assumption about equally likely gender probabilities is made, we still need to make additional assumptions before a solution can be given. This is because the sample space of the experiment consists of vectors of the form  $s_1, s_2, i$ , where  $s_1$  is the gender of the older child,  $s_2$  is the gender of the younger child, and  $i$  identifies the birth order of the child seen with the mother. As a result, to specify the probabilities of the events of the sample space, it is not enough to make assumptions only about the genders of the children; it is also necessary to assume something about the conditional probabilities as to which child is with the mother given the genders of the children. ■

**Example  
3n**

A bin contains 3 types of disposable flashlights. The probability that a type 1 flashlight will give more than 100 hours of use is .7, with the corresponding probabilities for type 2 and type 3 flashlights being .4 and .3, respectively. Suppose that 20 percent of the flashlights in the bin are type 1, 30 percent are type 2, and 50 percent are type 3.

- (a) What is the probability that a randomly chosen flashlight will give more than 100 hours of use?
- (b) Given that a flashlight lasted more than 100 hours, what is the conditional probability that it was a type  $j$  flashlight,  $j = 1, 2, 3$ ?

**Solution** (a) Let  $A$  denote the event that the flashlight chosen will give more than 100 hours of use, and let  $F_j$  be the event that a type  $j$  flashlight is chosen,  $j = 1, 2, 3$ . To compute  $P(A)$ , we condition on the type of the flashlight, to obtain

$$\begin{aligned} P(A) &= P(A|F_1)P(F_1) + P(A|F_2)P(F_2) + P(A|F_3)P(F_3) \\ &= (.7)(.2) + (.4)(.3) + (.3)(.5) = .41 \end{aligned}$$

There is a 41 percent chance that the flashlight will last for more than 100 hours.

(b) The probability is obtained by using Bayes's formula:

$$\begin{aligned} P(F_j|A) &= \frac{P(AF_j)}{P(A)} \\ &= \frac{P(A|F_j)P(F_j)}{.41} \end{aligned}$$

Thus,

$$\begin{aligned} P(F_1|A) &= (.7)(.2)/.41 = 14/41 \\ P(F_2|A) &= (.4)(.3)/.41 = 12/41 \\ P(F_3|A) &= (.3)(.5)/.41 = 15/41 \end{aligned}$$

For instance, whereas the initial probability that a type 1 flashlight is chosen is only .2, the information that the flashlight has lasted more than 100 hours raises the probability of this event to  $14/41 \approx .341$ . ■

**Example  
3o**

A crime has been committed by a solitary individual, who left some DNA at the scene of the crime. Forensic scientists who studied the recovered DNA noted that only five strands could be identified and that each innocent person, independently, would have a probability of  $10^{-5}$  of having his or her DNA match on all five strands. The district attorney supposes that the perpetrator of the crime could be any of the 1 million residents of the town. Ten thousand of these residents have been released from prison within the past 10 years; consequently, a sample of their DNA is on file. Before any checking of the DNA file, the district attorney thinks that each of the 10,000 ex-criminals has probability  $\alpha$  of being guilty of the new crime, whereas each of the remaining 990,000 residents has probability  $\beta$ , where  $\alpha = c\beta$ . (That is, the district attorney supposes that each recently released convict is  $c$  times as likely to be the crime's perpetrator as is each town member who is not a recently released convict.) When the DNA that is analyzed is compared against the database of the 10,000 ex-convicts, it turns out that A. J. Jones is the only one whose DNA matches the profile. Assuming that the district attorney's estimate of the relationship between  $\alpha$  and  $\beta$  is accurate, what is the probability that A. J. is guilty?

**Solution** To begin, note that because probabilities must sum to 1, we have

$$1 = 10,000\alpha + 990,000\beta = (10,000c + 990,000)\beta$$

Thus,

$$\beta = \frac{1}{10,000c + 990,000}, \quad \alpha = \frac{c}{10,000c + 990,000}$$

Now, let  $G$  be the event that A. J. is guilty, and let  $M$  denote the event that A. J. is the only one of the 10,000 on file to have a match. Then,

$$\begin{aligned} P(G|M) &= \frac{P(GM)}{P(M)} \\ &= \frac{P(G)P(M|G)}{P(M|G)P(G) + P(M|G^c)P(G^c)} \end{aligned}$$

On the one hand, if A. J. is guilty, then he will be the only one to have a DNA match if none of the others on file have a match. Therefore,

$$P(M|G) = (1 - 10^{-5})^{9999}$$

On the other hand, if A. J. is innocent, then in order for him to be the only match, his DNA must match (which will occur with probability  $10^{-5}$ ), all others in the database must be innocent, and none of these others can have a match. Now, given that A. J. is innocent, the conditional probability that all the others in the database are also innocent is

$$\begin{aligned} P(\text{all others innocent} | AJ \text{ innocent}) &= \frac{P(\text{all in database innocent})}{P(AJ \text{ innocent})} \\ &= \frac{1 - 10,000\alpha}{1 - \alpha} \end{aligned}$$

Also, the conditional probability, given their innocence, that none of the others in the database will have a match is  $(1 - 10^{-5})^{9999}$ . Therefore,

$$P(M|G^c) = 10^{-5} \left( \frac{1 - 10,000\alpha}{1 - \alpha} \right) (1 - 10^{-5})^{9999}$$

Because  $P(G) = \alpha$ , the preceding formula gives

$$P(G|M) = \frac{\alpha}{\alpha + 10^{-5}(1 - 10,000\alpha)} = \frac{1}{.9 + \frac{10^{-5}}{\alpha}}$$

Thus, if the district attorney's initial thoughts were that an arbitrary ex-convict was 100 times more likely to have committed the crime than was a nonconvict (that is,  $c = 100$ ), then  $\alpha = \frac{1}{19,900}$  and

$$P(G|M) = \frac{1}{1.099} \approx 0.9099$$

If the district attorney initially thought that the appropriate ratio was  $c = 10$ , then  $\alpha = \frac{1}{109,000}$  and

$$P(G|M) = \frac{1}{1.99} \approx 0.5025$$

If the district attorney initially thought that the criminal was equally likely to be any of the members of the town ( $c = 1$ ), then  $\alpha = 10^{-6}$  and

$$P(G|M) = \frac{1}{10.9} \approx 0.0917$$

Thus, the probability ranges from approximately 9 percent when the district attorney's initial assumption is that all the members of the population have the same chance of being the perpetrator to approximately 91 percent when she assumes that each ex-convict is 100 times more likely to be the criminal than is a specified towns person who is not an ex-convict. ■

## 4 Independent Events

The previous examples in this chapter show that  $P(E|F)$ , the conditional probability of  $E$  given  $F$ , is not generally equal to  $P(E)$ , the unconditional probability of  $E$ . In other words, knowing that  $F$  has occurred generally changes the chances of  $E$ 's occurrence. In the special cases where  $P(E|F)$  does in fact equal  $P(E)$ , we say that  $E$  is independent of  $F$ . That is,  $E$  is independent of  $F$  if knowledge that  $F$  has occurred does not change the probability that  $E$  occurs.

Since  $P(E|F) = P(EF)/P(F)$ , it follows that  $E$  is independent of  $F$  if

$$P(EF) = P(E)P(F) \quad (4.1)$$

The fact that Equation (4.1) is symmetric in  $E$  and  $F$  shows that whenever  $E$  is independent of  $F$ ,  $F$  is also independent of  $E$ . We thus have the following definition.

### Definition

Two events  $E$  and  $F$  are said to be *independent* if Equation (4.1) holds. Two events  $E$  and  $F$  that are not independent are said to be *dependent*.

#### Example 4a

A card is selected at random from an ordinary deck of 52 playing cards. If  $E$  is the event that the selected card is an ace and  $F$  is the event that it is a spade, then  $E$  and  $F$  are independent. This follows because  $P(EF) = \frac{1}{52}$ , whereas  $P(E) = \frac{4}{52}$  and  $P(F) = \frac{13}{52}$ . ■

#### Example 4b

Two coins are flipped, and all 4 outcomes are assumed to be equally likely. If  $E$  is the event that the first coin lands on heads and  $F$  the event that the second lands on tails, then  $E$  and  $F$  are independent, since  $P(EF) = P(\{(H, T)\}) = \frac{1}{4}$ , whereas  $P(E) = P(\{(H, H), (H, T)\}) = \frac{1}{2}$  and  $P(F) = P(\{(H, T), (T, T)\}) = \frac{1}{2}$ . ■

#### Example 4c

Suppose that we toss 2 fair dice. Let  $E_1$  denote the event that the sum of the dice is 6 and  $F$  denote the event that the first die equals 4. Then

$$P(E_1F) = P(\{(4, 2)\}) = \frac{1}{36}$$

whereas

$$P(E_1)P(F) = \left(\frac{5}{36}\right)\left(\frac{1}{6}\right) = \frac{5}{216}$$

Hence,  $E_1$  and  $F$  are not independent. Intuitively, the reason for this is clear because if we are interested in the possibility of throwing a 6 (with 2 dice), we shall be quite happy if the first die lands on 4 (or, indeed, on any of the numbers 1, 2, 3, 4, and 5), for then we shall still have a possibility of getting a total of 6. If, however, the first die landed on 6, we would be unhappy because we would no longer have a chance of getting a total of 6. In other words, our chance of getting a total of 6 depends on the outcome of the first die; thus,  $E_1$  and  $F$  cannot be independent.

Now, suppose that we let  $E_2$  be the event that the sum of the dice equals 7. Is  $E_2$  independent of  $F$ ? The answer is yes, since

$$P(E_2F) = P(\{(4,3)\}) = \frac{1}{36}$$

whereas

$$P(E_2)P(F) = \left(\frac{1}{6}\right)\left(\frac{1}{6}\right) = \frac{1}{36}$$

We leave it for the reader to present the intuitive argument why the event that the sum of the dice equals 7 is independent of the outcome on the first die. ■

**Example  
4d**

If we let  $E$  denote the event that the next president is a Republican and  $F$  the event that there will be a major earthquake within the next year, then most people would probably be willing to assume that  $E$  and  $F$  are independent. However, there would probably be some controversy over whether it is reasonable to assume that  $E$  is independent of  $G$ , where  $G$  is the event that there will be a recession within two years after the election. ■

We now show that if  $E$  is independent of  $F$ , then  $E$  is also independent of  $F^c$ .

**Proposition 4.1** If  $E$  and  $F$  are independent, then so are  $E$  and  $F^c$ .

**Proof** Assume that  $E$  and  $F$  are independent. Since  $E = EF \cup EF^c$  and  $EF$  and  $EF^c$  are obviously mutually exclusive, we have

$$\begin{aligned} P(E) &= P(EF) + P(EF^c) \\ &= P(E)P(F) + P(EF^c) \end{aligned}$$

or, equivalently,

$$\begin{aligned} P(EF^c) &= P(E)[1 - P(F)] \\ &= P(E)P(F^c) \end{aligned}$$

and the result is proved. □

Thus, if  $E$  is independent of  $F$ , then the probability of  $E$ 's occurrence is unchanged by information as to whether or not  $F$  has occurred.

Suppose now that  $E$  is independent of  $F$  and is also independent of  $G$ . Is  $E$  then necessarily independent of  $FG$ ? The answer, somewhat surprisingly, is no, as the following example demonstrates.

**Example  
4e**

Two fair dice are thrown. Let  $E$  denote the event that the sum of the dice is 7. Let  $F$  denote the event that the first die equals 4 and  $G$  denote the event that the second die equals 3. From Example 4c, we know that  $E$  is independent of  $F$ , and the same reasoning as applied there shows that  $E$  is also independent of  $G$ ; but clearly,  $E$  is not independent of  $FG$  [since  $P(E|FG) = 1$ ]. ■

It would appear to follow from Example 4e that an appropriate definition of the independence of three events  $E$ ,  $F$ , and  $G$  would have to go further than merely assuming that all of the  $\binom{3}{2}$  pairs of events are independent. We are thus led to the following definition.

**Definition**

Three events  $E$ ,  $F$ , and  $G$  are said to be independent if

$$P(EFG) = P(E)P(F)P(G)$$

$$P(EF) = P(E)P(F)$$

$$P(EG) = P(E)P(G)$$

$$P(FG) = P(F)P(G)$$

Note that if  $E$ ,  $F$ , and  $G$  are independent, then  $E$  will be independent of any event formed from  $F$  and  $G$ . For instance,  $E$  is independent of  $F \cup G$ , since

$$\begin{aligned} P[E(F \cup G)] &= P(EF \cup EG) \\ &= P(EF) + P(EG) - P(EFG) \\ &= P(E)P(F) + P(E)P(G) - P(E)P(FG) \\ &= P(E)[P(F) + P(G) - P(FG)] \\ &= P(E)P(F \cup G) \end{aligned}$$

Of course, we may also extend the definition of independence to more than three events. The events  $E_1, E_2, \dots, E_n$  are said to be independent if for every subset  $E_{1'}, E_{2'}, \dots, E_{r'}, r \leq n$  of these events,

$$P(E_{1'}E_{2'} \cdots E_{r'}) = P(E_{1'})P(E_{2'}) \cdots P(E_{r'})$$

Finally, we define an infinite set of events to be independent if every finite subset of those events is independent.

Sometimes, a probability experiment under consideration consists of performing a sequence of subexperiments. For instance, if the experiment consists of continually tossing a coin, we may think of each toss as being a subexperiment. In many cases, it is reasonable to assume that the outcomes of any group of the subexperiments have no effect on the probabilities of the outcomes of the other subexperiments. If such is the case, we say that the subexperiments are independent. More formally, we say that the subexperiments are independent if  $E_1, E_2, \dots, E_n, \dots$  is necessarily an independent sequence of events whenever  $E_i$  is an event whose occurrence is completely determined by the outcome of the  $i$ th subexperiment.

If each subexperiment has the same set of possible outcomes, then the subexperiments are often called *trials*.

**Example  
4f**

An infinite sequence of independent trials is to be performed. Each trial results in a success with probability  $p$  and a failure with probability  $1 - p$ . What is the probability that

- (a) at least 1 success occurs in the first  $n$  trials;
- (b) exactly  $k$  successes occur in the first  $n$  trials;
- (c) all trials result in successes?

**Solution** In order to determine the probability of at least 1 success in the first  $n$  trials, it is easiest to compute first the probability of the complementary event: that of no successes in the first  $n$  trials. If we let  $E_i$  denote the event of a failure on the  $i$ th trial, then the probability of no successes is, by independence,

$$P(E_1 E_2 \cdots E_n) = P(E_1)P(E_2) \cdots P(E_n) = (1 - p)^n$$

Hence, the answer to part (a) is  $1 - (1 - p)^n$ .

To compute the answer to part (b), consider any particular sequence of the first  $n$  outcomes containing  $k$  successes and  $n - k$  failures. Each one of these sequences will, by the assumed independence of trials, occur with probability  $p^k(1 - p)^{n-k}$ .

Since there are  $\binom{n}{k}$  such sequences [there are  $n!/k!(n - k)!$  permutations of  $k$  successes and  $n - k$  failures], the desired probability in part (b) is

$$P\{\text{exactly } k \text{ successes}\} = \binom{n}{k} p^k (1 - p)^{n-k}$$

To answer part (c), we note that, by part (a), the probability of the first  $n$  trials all resulting in success is given by

$$P(E_1^c E_2^c \cdots E_n^c) = p^n$$

Thus, using the continuity property of probabilities, we see that the desired probability is given by

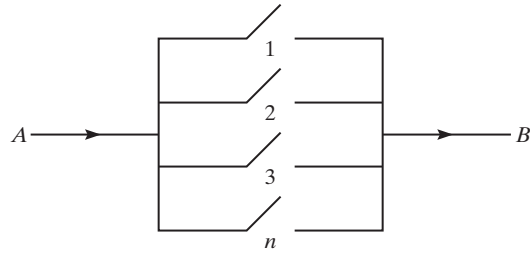
$$\begin{aligned} P\left(\bigcap_{i=1}^{\infty} E_i^c\right) &= P\left(\lim_{n \rightarrow \infty} \bigcap_{i=1}^n E_i^c\right) \\ &= \lim_{n \rightarrow \infty} P\left(\bigcap_{i=1}^n E_i^c\right) \\ &= \lim_n p^n = \begin{cases} 0 & \text{if } p < 1 \\ 1 & \text{if } p = 1 \end{cases} \quad \blacksquare \end{aligned}$$

**Example 4g**

A system composed of  $n$  separate components is said to be a parallel system if it functions when at least one of the components functions. (See Figure 2.) For such a system, if component  $i$ , which is independent of the other components, functions with probability  $p_i, i = 1, \dots, n$ , what is the probability that the system functions?

**Solution** Let  $A_i$  denote the event that component  $i$  functions. Then,

$$\begin{aligned} P\{\text{system functions}\} &= 1 - P\{\text{system does not function}\} \\ &= 1 - P\{\text{all components do not function}\} \\ &= 1 - P\left(\bigcap_i A_i^c\right) \\ &= 1 - \prod_{i=1}^n (1 - p_i) \quad \text{by independence} \quad \blacksquare \end{aligned}$$



**Figure 2** Parallel System: Functions if Current Flows from  $A$  to  $B$ .

**Example  
4h**

Independent trials consisting of rolling a pair of fair dice are performed. What is the probability that an outcome of 5 appears before an outcome of 7 when the outcome of a roll is the sum of the dice?

**Solution** If we let  $E_n$  denote the event that no 5 or 7 appears on the first  $n - 1$  trials and a 5 appears on the  $n$ th trial, then the desired probability is

$$P\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} P(E_n)$$

Now, since  $P\{5 \text{ on any trial}\} = \frac{4}{36}$  and  $P\{7 \text{ on any trial}\} = \frac{6}{36}$ , we obtain, by the independence of trials,

$$P(E_n) = \left(1 - \frac{10}{36}\right)^{n-1} \frac{4}{36}$$

Thus,

$$\begin{aligned} P\left(\bigcup_{n=1}^{\infty} E_n\right) &= \frac{1}{9} \sum_{n=1}^{\infty} \left(\frac{13}{18}\right)^{n-1} \\ &= \frac{1}{9} \frac{1}{1 - \frac{13}{18}} \\ &= \frac{2}{5} \end{aligned}$$

This result could also have been obtained by the use of conditional probabilities. If we let  $E$  be the event that a 5 occurs before a 7, then we can obtain the desired probability,  $P(E)$ , by conditioning on the outcome of the first trial, as follows: Let  $F$  be the event that the first trial results in a 5, let  $G$  be the event that it results in a 7, and let  $H$  be the event that the first trial results in neither a 5 nor a 7. Then, conditioning on which one of these events occurs gives

$$P(E) = P(E|F)P(F) + P(E|G)P(G) + P(E|H)P(H)$$

However,

$$\begin{aligned} P(E|F) &= 1 \\ P(E|G) &= 0 \\ P(E|H) &= P(E) \end{aligned}$$

The first two equalities are obvious. The third follows because if the first outcome results in neither a 5 nor a 7, then at that point the situation is exactly as it was when

the problem first started—namely, the experimenter will continually roll a pair of fair dice until either a 5 or 7 appears. Furthermore, the trials are independent; therefore, the outcome of the first trial will have no effect on subsequent rolls of the dice. Since  $P(F) = \frac{4}{36}$ ,  $P(G) = \frac{6}{36}$ , and  $P(H) = \frac{26}{36}$ , it follows that

$$P(E) = \frac{1}{9} + P(E)\frac{13}{18}$$

or

$$P(E) = \frac{2}{5}$$

The reader should note that the answer is quite intuitive. That is, because a 5 occurs on any roll with probability  $\frac{4}{36}$  and a 7 with probability  $\frac{6}{36}$ , it seems intuitive that the odds that a 5 appears before a 7 should be 6 to 4 against. The probability should then be  $\frac{4}{10}$ , as indeed it is.

The same argument shows that if  $E$  and  $F$  are mutually exclusive events of an experiment, then, when independent trials of the experiment are performed, the event  $E$  will occur before the event  $F$  with probability

$$\frac{P(E)}{P(E) + P(F)}$$

■

**Example**  
**4i**

Suppose there are  $n$  types of coupons and that each new coupon collected is, independent of previous selections, a type  $i$  coupon with probability  $p_i$ ,  $\sum_{i=1}^n p_i = 1$ . Suppose  $k$  coupons are to be collected. If  $A_i$  is the event that there is at least one type  $i$  coupon among those collected, then, for  $i \neq j$ , find

- (a)  $P(A_i)$
- (b)  $P(A_i \cup A_j)$
- (c)  $P(A_i|A_j)$

**Solution**

$$\begin{aligned} P(A_i) &= 1 - P(A_i^c) \\ &= 1 - P\{\text{no coupon is type } i\} \\ &= 1 - (1 - p_i)^k \end{aligned}$$

where the preceding used that each coupon is, independently, not of type  $i$  with probability  $1 - p_i$ . Similarly,

$$\begin{aligned} P(A_i \cup A_j) &= 1 - P((A_i \cup A_j)^c) \\ &= 1 - P\{\text{no coupon is either type } i \text{ or type } j\} \\ &= 1 - (1 - p_i - p_j)^k \end{aligned}$$

where the preceding used that each coupon is, independently, neither of type  $i$  nor type  $j$  with probability  $1 - p_i - p_j$ .

To determine  $P(A_i|A_j)$ , we will use the identity

$$P(A_i \cup A_j) = P(A_i) + P(A_j) - P(A_i A_j)$$

which, in conjunction with parts (a) and (b), yields

$$\begin{aligned} P(A_i A_j) &= 1 - (1 - p_i)^k + 1 - (1 - p_j)^k - [1 - (1 - p_i - p_j)^k] \\ &= 1 - (1 - p_i)^k - (1 - p_j)^k + (1 - p_i - p_j)^k \end{aligned}$$

Consequently,

$$P(A_i|A_j) = \frac{P(A_i A_j)}{P(A_j)} = \frac{1 - (1 - p_i)^k - (1 - p_j)^k + (1 - p_i - p_j)^k}{1 - (1 - p_j)^k} \quad \blacksquare$$

The next example presents a problem that occupies an honored place in the history of probability theory. This is the famous *problem of the points*. In general terms, the problem is this: Two players put up stakes and play some game, with the stakes to go to the winner of the game. An interruption requires them to stop before either has won and when each has some sort of a “partial score.” How should the stakes be divided?

This problem was posed to the French mathematician Blaise Pascal in 1654 by the Chevalier de Méré, who was a professional gambler at that time. In attacking the problem, Pascal introduced the important idea that the proportion of the prize deserved by the competitors should depend on their respective probabilities of winning if the game were to be continued at that point. Pascal worked out some special cases and, more importantly, initiated a correspondence with the famous Frenchman Pierre de Fermat, who had a reputation as a great mathematician. The resulting exchange of letters not only led to a complete solution to the problem of the points, but also laid the framework for the solution to many other problems connected with games of chance. This celebrated correspondence, considered by some as the birth date of probability theory, was also important in stimulating interest in probability among the mathematicians in Europe, for Pascal and Fermat were both recognized as being among the foremost mathematicians of the time. For instance, within a short time of their correspondence, the young Dutch mathematician Christiaan Huygens came to Paris to discuss these problems and solutions, and interest and activity in this new field grew rapidly.

**Example**  
4j

**The problem of the points**

Independent trials resulting in a success with probability  $p$  and a failure with probability  $1 - p$  are performed. What is the probability that  $n$  successes occur before  $m$  failures? If we think of  $A$  and  $B$  as playing a game such that  $A$  gains 1 point when a success occurs and  $B$  gains 1 point when a failure occurs, then the desired probability is the probability that  $A$  would win if the game were to be continued in a position where  $A$  needed  $n$  and  $B$  needed  $m$  more points to win.

**Solution** We shall present two solutions. The first comes from Pascal and the second from Fermat.

Let us denote by  $P_{n,m}$  the probability that  $n$  successes occur before  $m$  failures. By conditioning on the outcome of the first trial, we obtain

$$P_{n,m} = pP_{n-1,m} + (1 - p)P_{n,m-1} \quad n \geq 1, m \geq 1$$

(Why? Reason it out.) Using the obvious boundary conditions  $P_{n,0} = 0, P_{0,m} = 1$ , we can solve these equations for  $P_{n,m}$ . Rather than go through the tedious details, let us instead consider Fermat’s solution.

Fermat argued that in order for  $n$  successes to occur before  $m$  failures, it is necessary and sufficient that there be at least  $n$  successes in the first  $m + n - 1$  trials. (Even if the game were to end before a total of  $m + n - 1$  trials were completed, we could still imagine that the necessary additional trials were performed.) This is true, for if there are at least  $n$  successes in the first  $m + n - 1$  trials, there could be at most  $m - 1$  failures in those  $m + n - 1$  trials; thus,  $n$  successes would occur before  $m$  failures. If, however, there were fewer than  $n$  successes in the first  $m + n - 1$

trials, there would have to be at least  $m$  failures in that same number of trials; thus,  $n$  successes would not occur before  $m$  failures.

Hence, since, as shown in Example 4f, the probability of exactly  $k$  successes in  $m + n - 1$  trials is  $\binom{m+n-1}{k} p^k (1-p)^{m+n-1-k}$ , it follows that the desired probability of  $n$  successes before  $m$  failures is

$$P_{n,m} = \sum_{k=n}^{m+n-1} \binom{m+n-1}{k} p^k (1-p)^{m+n-1-k} \quad \blacksquare$$

The following example gives another instance where determining the probability that a player wins a match is made easier by assuming that the play continues even after the match winner has been determined.

#### Example 4k

##### Service protocol in a serve and rally game

Consider a serve and rally match (such as volleyball, badminton, or squash) between two players,  $A$  and  $B$ . The match consists of a sequence of rallies, with each rally beginning with a serve by one of the players and continuing until one of the players has won the rally. The winner of the rally receives a point, and the match ends when one of the players has won a total of  $n$  points, with that player being declared the winner of the match. Suppose whenever a rally begins with  $A$  as the server, that  $A$  wins that rally with probability  $p_A$  and that  $B$  wins it with probability  $q_A = 1 - p_A$ , and that a rally that begins with  $B$  as the server is won by  $A$  with probability  $p_B$  and by  $B$  with probability  $q_B = 1 - p_B$ . Player  $A$  is to be the initial server. There are two possible server protocols that are under consideration: “winner serves,” which means that the winner of a rally is the server for the next rally, or “alternating serve,” which means that the server alternates from rally to rally, so that no two consecutive rallies have the same server. Thus, for instance, if  $n = 3$ , then the successive servers under the “winner serves” protocol would be  $A, B, A, A$  if  $A$  wins the first point, then  $B$  the next, then  $A$  wins the next two. On the other hand, the sequence of servers under the “alternating serve” protocol will always be  $A, B, A, B, A, \dots$  until the match winner is decided. If you were player  $A$ , which protocol would you prefer?

**Solution** Surprisingly, it turns out that it makes no difference, in that the probability that  $A$  is the match winner is the same under either protocol. To show that this is the case, it is advantageous to suppose that the players continue to play until a total of  $2n - 1$  rallies have been completed. The first player to win  $n$  rallies would then be the one who has won at least  $n$  of the  $2n - 1$  rallies. To begin, note that if the alternating serve protocol is being used, then player  $A$  will serve exactly  $n$  times and player  $B$  will serve exactly  $n - 1$  times in the  $2n - 1$  rallies.

Now consider the winner serve protocol, again assuming that the players continue to play until  $2n - 1$  rallies have been completed. Because it makes no difference who serves the “extra rallies” after the match winner has been decided, suppose that at the point at which the match has been decided (because one of the players has won  $n$  points), the remainder (if there are any) of the  $2n - 1$  rallies are all served by the player who lost the match. Note that this modified service protocol does not change the fact that the winner of the match will still be the player who wins at least  $n$  of the  $2n - 1$  rallies. We claim that under this modified service protocol,  $A$  will always serve  $n$  times and  $B$  will always serve  $n - 1$  times. Two cases show this.

*Case 1: A wins the match.*

Because  $A$  serves first, it follows that  $A$ ’s second serve will immediately follow  $A$ ’s first point;  $A$ ’s third serve will immediately follow  $A$ ’s second point; and, in particular,

$A$ 's  $n$ th serve will immediately follow  $A$ 's  $(n - 1)$  point. But this will be the last serve of  $A$  before the match result is decided. This is so because either  $A$  will win the point on that serve and so have  $n$  points, or  $A$  will lose the point and so the serve will switch to, and remain with,  $B$  until  $A$  wins point number  $n$ . Thus, provided that  $A$  wins the match, it follows that  $A$  would have served a total of  $n$  times at the moment the match is decided. Because, by the modified service protocol,  $A$  will never again serve, it follows in this case that  $A$  serves exactly  $n$  times.

*Case 2:  $B$  wins the match.*

Because  $A$  serves first,  $B$ 's first serve will come immediately after  $B$ 's first point;  $B$ 's second serve will come immediately after  $B$ 's second point; and, in particular,  $B$ 's  $(n - 1)$  serve will come immediately after  $B$ 's  $(n - 1)$  point. But that will be the last serve of  $B$  before the match is decided because either  $B$  will win the point on that serve and so have  $n$  points, or  $B$  will lose the point and so the serve will switch to, and remain with,  $A$  until  $B$  wins point number  $n$ . Thus, provided that  $B$  wins the match, we see that  $B$  would have served a total of  $n - 1$  times at the moment the match is decided. Because, by the modified service protocol,  $B$  will never again serve, it follows in this case that  $B$  serves exactly  $n - 1$  times, and, as there are a total of  $2n - 1$  rallies, that  $A$  serves exactly  $n$  times.

Thus, we see that under either protocol,  $A$  will always serve  $n$  times and  $B$  will serve  $n - 1$  times and the winner of the match will be the one who wins at least  $n$  points. But since  $A$  wins each rally that he serves with probability  $p_A$  and wins each rally that  $B$  serves with probability  $p_B$  it follows that the probability that  $A$  is the match winner is, under either protocol, equal to the probability that there are at least  $n$  successes in  $2n - 1$  independent trials, when  $n$  of these trials result in a success with probability  $p_A$  and the other  $n - 1$  trials result in a success with probability  $p_B$ . Consequently, the win probabilities for both protocols are the same. ■

Our next two examples deal with gambling problems, with the first having a surprisingly elegant analysis.\*

**Example**  
**41**

Suppose that initially there are  $r$  players, with player  $i$  having  $n_i$  units,  $n_i > 0$ ,  $i = 1, \dots, r$ . At each stage, two of the players are chosen to play a game, with the winner of the game receiving 1 unit from the loser. Any player whose fortune drops to 0 is eliminated, and this continues until a single player has all  $n = \sum_{i=1}^r n_i$  units, with that player designated as the victor. Assuming that the results of successive games are independent and that each game is equally likely to be won by either of its two players, find  $P_i$ , the probability that player  $i$  is the victor.

**Solution** To begin, suppose that there are  $n$  players, with each player initially having 1 unit. Consider player  $i$ . Each stage she plays will be equally likely to result in her either winning or losing 1 unit, with the results from each stage being independent. In addition, she will continue to play stages until her fortune becomes either 0 or  $n$ . Because this is the same for all  $n$  players, it follows that each player has the same chance of being the victor, implying that each player has probability  $1/n$  of being the victor. Now, suppose these  $n$  players are divided into  $r$  teams, with team  $i$  containing  $n_i$  players,  $i = 1, \dots, r$ . Then, the probability that the victor is a member of team  $i$  is  $n_i/n$ . But because

- (a) team  $i$  initially has a total fortune of  $n_i$  units,  $i = 1, \dots, r$ , and
- (b) each game played by members of different teams is equally likely to be won by either player and results in the fortune of members of the winning team increasing by 1 and the fortune of the members of the losing team decreasing by 1,

\*The remainder of this section should be considered optional.

it is easy to see that the probability that the victor is from team  $i$  is exactly the probability we desire. Thus,  $P_i = n_i/n$ . Interestingly, our argument shows that this result does *not* depend on how the players in each stage are chosen. ■

In the *gambler's ruin problem*, there are only 2 gamblers, but they are not assumed to be of equal skill.

**Example**  
4m

**The gambler's ruin problem**

Two gamblers,  $A$  and  $B$ , bet on the outcomes of successive flips of a coin. On each flip, if the coin comes up heads,  $A$  collects 1 unit from  $B$ , whereas if it comes up tails,  $A$  pays 1 unit to  $B$ . They continue to do this until one of them runs out of money. If it is assumed that the successive flips of the coin are independent and each flip results in a head with probability  $p$ , what is the probability that  $A$  ends up with all the money if he starts with  $i$  units and  $B$  starts with  $N - i$  units?

**Solution** Let  $E$  denote the event that  $A$  ends up with all the money when he starts with  $i$  and  $B$  starts with  $N - i$ , and to make clear the dependence on the initial fortune of  $A$ , let  $P_i = P(E)$ . We shall obtain an expression for  $P(E)$  by conditioning on the outcome of the first flip as follows: Let  $H$  denote the event that the first flip lands on heads; then

$$\begin{aligned} P_i = P(E) &= P(E|H)P(H) + P(E|H^c)P(H^c) \\ &= pP(E|H) + (1 - p)P(E|H^c) \end{aligned}$$

Now, given that the first flip lands on heads, the situation after the first bet is that  $A$  has  $i + 1$  units and  $B$  has  $N - (i + 1)$ . Since the successive flips are assumed to be independent with a common probability  $p$  of heads, it follows that from that point on,  $A$ 's probability of winning all the money is exactly the same as if the game were just starting with  $A$  having an initial fortune of  $i + 1$  and  $B$  having an initial fortune of  $N - (i + 1)$ . Therefore,

$$P(E|H) = P_{i+1}$$

and similarly,

$$P(E|H^c) = P_{i-1}$$

Hence, letting  $q = 1 - p$ , we obtain

$$P_i = pP_{i+1} + qP_{i-1} \quad i = 1, 2, \dots, N - 1 \quad (4.2)$$

By making use of the obvious boundary conditions  $P_0 = 0$  and  $P_N = 1$ , we shall now solve Equation (4.2). Since  $p + q = 1$ , these equations are equivalent to

$$pP_i + qP_i = pP_{i+1} + qP_{i-1}$$

or

$$P_{i+1} - P_i = \frac{q}{p}(P_i - P_{i-1}) \quad i = 1, 2, \dots, N - 1 \quad (4.3)$$

Hence, since  $P_0 = 0$ , we obtain, from Equation (4.3),

$$\begin{aligned}
 P_2 - P_1 &= \frac{q}{p}(P_1 - P_0) = \frac{q}{p}P_1 \\
 P_3 - P_2 &= \frac{q}{p}(P_2 - P_1) = \left(\frac{q}{p}\right)^2 P_1 \\
 &\vdots \\
 P_i - P_{i-1} &= \frac{q}{p}(P_{i-1} - P_{i-2}) = \left(\frac{q}{p}\right)^{i-1} P_1 \\
 &\vdots \\
 P_N - P_{N-1} &= \frac{q}{p}(P_{N-1} - P_{N-2}) = \left(\frac{q}{p}\right)^{N-1} P_1
 \end{aligned} \tag{4.4}$$

Adding the first  $i - 1$  equations of (4.4) yields

$$P_i - P_1 = P_1 \left[ \left(\frac{q}{p}\right) + \left(\frac{q}{p}\right)^2 + \cdots + \left(\frac{q}{p}\right)^{i-1} \right]$$

or

$$P_i = \begin{cases} \frac{1 - (q/p)^i}{1 - (q/p)} P_1 & \text{if } \frac{q}{p} \neq 1 \\ iP_1 & \text{if } \frac{q}{p} = 1 \end{cases}$$

Using the fact that  $P_N = 1$ , we obtain

$$P_1 = \begin{cases} \frac{1 - (q/p)}{1 - (q/p)^N} & \text{if } p \neq \frac{1}{2} \\ \frac{1}{N} & \text{if } p = \frac{1}{2} \end{cases}$$

Hence,

$$P_i = \begin{cases} \frac{1 - (q/p)^i}{1 - (q/p)^N} & \text{if } p \neq \frac{1}{2} \\ \frac{i}{N} & \text{if } p = \frac{1}{2} \end{cases} \tag{4.5}$$

Let  $Q_i$  denote the probability that  $B$  winds up with all the money when  $A$  starts with  $i$  and  $B$  starts with  $N - i$ . Then, by symmetry to the situation described, and on replacing  $p$  by  $q$  and  $i$  by  $N - i$ , it follows that

$$Q_i = \begin{cases} \frac{1 - (p/q)^{N-i}}{1 - (p/q)^N} & \text{if } q \neq \frac{1}{2} \\ \frac{N - i}{N} & \text{if } q = \frac{1}{2} \end{cases}$$

Moreover, since  $q = \frac{1}{2}$  is equivalent to  $p = \frac{1}{2}$ , we have, when  $q \neq \frac{1}{2}$ ,

$$\begin{aligned} P_i + Q_i &= \frac{1 - (q/p)^i}{1 - (q/p)^N} + \frac{1 - (p/q)^{N-i}}{1 - (p/q)^N} \\ &= \frac{p^N - p^N(q/p)^i}{p^N - q^N} + \frac{q^N - q^N(p/q)^{N-i}}{q^N - p^N} \\ &= \frac{p^N - p^{N-i}q^i - q^N + q^i p^{N-i}}{p^N - q^N} \\ &= 1 \end{aligned}$$

This result also holds when  $p = q = \frac{1}{2}$ , so

$$P_i + Q_i = 1$$

In words, this equation states that with probability 1, either  $A$  or  $B$  will wind up with all of the money; in other words, the probability that the game continues indefinitely with  $A$ 's fortune always being between 1 and  $N - 1$  is zero. (The reader must be careful because, a priori, there are three possible outcomes of this gambling game, not two: Either  $A$  wins, or  $B$  wins, or the game goes on forever with nobody winning. We have just shown that this last event has probability 0.)

As a numerical illustration of the preceding result, if  $A$  were to start with 5 units and  $B$  with 10, then the probability of  $A$ 's winning would be  $\frac{1}{3}$  if  $p$  were  $\frac{1}{2}$ , whereas it would jump to

$$\frac{1 - \left(\frac{2}{3}\right)^5}{1 - \left(\frac{2}{3}\right)^{15}} \approx .87$$

if  $p$  were .6.

A special case of the gambler's ruin problem, which is also known as the problem of *duration of play*, was proposed to Huygens by Fermat in 1657. The version Huygens proposed, which he himself solved, was that  $A$  and  $B$  have 12 coins each. They play for these coins in a game with 3 dice as follows: Whenever 11 is thrown (by either—it makes no difference who rolls the dice),  $A$  gives a coin to  $B$ . Whenever 14 is thrown,  $B$  gives a coin to  $A$ . The person who first wins all the coins wins the game. Since  $P\{\text{roll } 11\} = \frac{27}{216}$  and  $P\{\text{roll } 14\} = \frac{15}{216}$ , we see from Example 4h that, for  $A$ , this is just the gambler's ruin problem with  $p = \frac{15}{42}$ ,  $i = 12$ , and  $N = 24$ . The general form of the gambler's ruin problem was solved by the mathematician James Bernoulli and published 8 years after his death in 1713.

For an application of the gambler's ruin problem to drug testing, suppose that two new drugs have been developed for treating a certain disease. Drug  $i$  has a cure rate  $P_i$ ,  $i = 1, 2$ , in the sense that each patient treated with drug  $i$  will be cured with probability  $P_i$ . These cure rates are, however, not known, and we are interested in finding a method for deciding whether  $P_1 > P_2$  or  $P_2 > P_1$ . To decide on one of these alternatives, consider the following test: Pairs of patients are to be treated sequentially, with one member of the pair receiving drug 1 and the other drug 2. The results for each pair are determined, and the testing stops when the cumulative number of cures from one of the drugs exceeds the cumulative number of cures from the other by some fixed, predetermined number. More formally, let

$$\begin{aligned} X_j &= \begin{cases} 1 & \text{if the patient in the } j\text{th pair that receives drug 1 is cured} \\ 0 & \text{otherwise} \end{cases} \\ Y_j &= \begin{cases} 1 & \text{if the patient in the } j\text{th pair that receives drug 2 is cured} \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

For a predetermined positive integer  $M$ , the test stops after pair  $N$ , where  $N$  is the first value of  $n$  such that either

$$X_1 + \cdots + X_n - (Y_1 + \cdots + Y_n) = M$$

or

$$X_1 + \cdots + X_n - (Y_1 + \cdots + Y_n) = -M$$

In the former case, we assert that  $P_1 > P_2$  and in the latter that  $P_2 > P_1$ .

In order to help ascertain whether the foregoing is a good test, one thing we would like to know is the probability that it leads to an incorrect decision. That is, for given  $P_1$  and  $P_2$ , where  $P_1 > P_2$ , what is the probability that the test will incorrectly assert that  $P_2 > P_1$ ? To determine this probability, note that after each pair is checked, the cumulative difference of cures using drug 1 versus drug 2 will go up by 1 with probability  $P_1(1 - P_2)$ —since this is the probability that drug 1 leads to a cure and drug 2 does not—or go down by 1 with probability  $(1 - P_1)P_2$ , or remain the same with probability  $P_1P_2 + (1 - P_1)(1 - P_2)$ . Hence, if we consider only those pairs in which the cumulative difference changes, then the difference will go up by 1 with probability

$$\begin{aligned} P &= P\{\text{up 1} | \text{up 1 or down 1}\} \\ &= \frac{P_1(1 - P_2)}{P_1(1 - P_2) + (1 - P_1)P_2} \end{aligned}$$

and down by 1 with probability

$$1 - P = \frac{P_2(1 - P_1)}{P_1(1 - P_2) + (1 - P_1)P_2}$$

Thus, the probability that the test will assert that  $P_2 > P_1$  is equal to the probability that a gambler who wins each (one-unit) bet with probability  $P$  will go down  $M$  before going up  $M$ . But Equation (4.5), with  $i = M, N = 2M$ , shows that this probability is given by

$$\begin{aligned} &P\{\text{test asserts that } P_2 > P_1\} \\ &= 1 - \frac{1 - \left(\frac{1 - P}{P}\right)^M}{1 - \left(\frac{1 - P}{P}\right)^{2M}} \\ &= 1 - \frac{1}{1 + \left(\frac{1 - P}{P}\right)^M} \\ &= \frac{1}{1 + \gamma^M} \end{aligned}$$

where

$$\gamma = \frac{P}{1 - P} = \frac{P_1(1 - P_2)}{P_2(1 - P_1)}$$

For instance, if  $P_1 = .6$  and  $P_2 = .4$ , then the probability of an incorrect decision is .017 when  $M = 5$  and reduces to .0003 when  $M = 10$ . ■

Suppose that we are presented with a set of elements and we want to determine whether at least one member of the set has a certain property. We can attack this question probabilistically by randomly choosing an element of the set in such a way that each element has a positive probability of being selected. Then the original

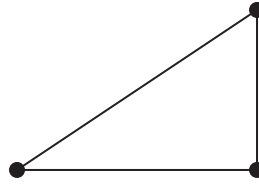
question can be answered by a consideration of the probability that the randomly selected element does not have the property of interest. If this probability is equal to 1, then none of the elements of the set has the property; if it is less than 1, then at least one element of the set has the property.

The final example of this section illustrates this technique.

**Example**  
**4n**

The complete graph having  $n$  vertices is defined to be a set of  $n$  points (called vertices) in the plane and the  $\binom{n}{2}$  lines (called edges) connecting each pair of vertices.

The complete graph having 3 vertices is shown in Figure 3. Suppose now that each edge in a complete graph having  $n$  vertices is to be colored either red or blue. For a fixed integer  $k$ , a question of interest is, Is there a way of coloring the edges so that no set of  $k$  vertices has all of its  $\binom{k}{2}$  connecting edges the same color? It can be shown by a probabilistic argument that if  $n$  is not too large, then the answer is yes.



**Figure 3**

The argument runs as follows: Suppose that each edge is, independently, equally likely to be colored either red or blue. That is, each edge is red with probability  $\frac{1}{2}$ . Number the  $\binom{n}{k}$  sets of  $k$  vertices and define the events  $E_i, i = 1, \dots, \binom{n}{k}$  as follows:

$$E_i = \{\text{all of the connecting edges of the } i\text{th set of } k \text{ vertices are the same color}\}$$

Now, since each of the  $\binom{k}{2}$  connecting edges of a set of  $k$  vertices is equally likely to be either red or blue, it follows that the probability that they are all the same color is

$$P(E_i) = 2 \left( \frac{1}{2} \right)^{k(k-1)/2}$$

Therefore, because

$$P \left( \bigcup_i E_i \right) \leq \sum_i P(E_i) \quad (\text{Boole's inequality})$$

we find that  $P \left( \bigcup_i E_i \right)$ , the probability that there is a set of  $k$  vertices all of whose connecting edges are similarly colored, satisfies

$$P \left( \bigcup_i E_i \right) \leq \binom{n}{k} \left( \frac{1}{2} \right)^{k(k-1)/2-1}$$

Hence, if

$$\binom{n}{k} \left(\frac{1}{2}\right)^{k(k-1)/2-1} < 1$$

or, equivalently, if

$$\binom{n}{k} < 2^{k(k-1)/2-1}$$

then the probability that at least one of the  $\binom{n}{k}$  sets of  $k$  vertices has all of its connecting edges the same color is less than 1. Consequently, under the preceding condition on  $n$  and  $k$ , it follows that there is a positive probability that no set of  $k$  vertices has all of its connecting edges the same color. But this conclusion implies that there is at least one way of coloring the edges for which no set of  $k$  vertices has all of its connecting edges the same color. ■

**Remarks** (a) Whereas the preceding argument established a condition on  $n$  and  $k$  that guarantees the existence of a coloring scheme satisfying the desired property, it gives no information about how to obtain such a scheme (although one possibility would be simply to choose the colors at random, check to see if the resulting coloring satisfies the property, and repeat the procedure until it does).

(b) The method of introducing probability into a problem whose statement is purely deterministic has been called the *probabilistic method*.<sup>†</sup> Other examples of this method are given in Theoretical Exercise 24.

## 5 $P(\cdot | F)$ Is a Probability

Conditional probabilities satisfy all of the properties of ordinary probabilities, as is proved by Proposition 5.1, which shows that  $P(E|F)$  satisfies the three axioms of a probability.

**Proposition 5.1** (a)  $0 \leq P(E|F) \leq 1$ .  
 (b)  $P(S|F) = 1$ .  
 (c) If  $E_i, i = 1, 2, \dots$ , are mutually exclusive events, then

$$P\left(\bigcup_{i=1}^{\infty} E_i | F\right) = \sum_{i=1}^{\infty} P(E_i | F)$$

**Proof** To prove part (a), we must show that  $0 \leq P(EF)/P(F) \leq 1$ . The left-side inequality is obvious, whereas the right side follows because  $EF \subset F$ , which implies that  $P(EF) \leq P(F)$ . Part (b) follows because

$$P(S|F) = \frac{P(SF)}{P(F)} = \frac{P(F)}{P(F)} = 1$$

<sup>†</sup>See N. Alon, J. Spencer, and P. Erdos, *The Probabilistic Method* (New York: John Wiley & Sons, Inc., 1992).

Part (c) follows from

$$\begin{aligned}
 P\left(\bigcup_{i=1}^{\infty} E_i | F\right) &= \frac{P\left(\left(\bigcup_{i=1}^{\infty} E_i\right) F\right)}{P(F)} \\
 &= \frac{P\left(\bigcup_1^{\infty} E_i F\right)}{P(F)} \quad \text{since} \quad \left(\bigcup_1^{\infty} E_i\right) F = \bigcup_1^{\infty} E_i F \\
 &= \frac{\sum_1^{\infty} P(E_i F)}{P(F)} \\
 &= \sum_1^{\infty} P(E_i | F)
 \end{aligned}$$

where the next-to-last equality follows because  $E_i E_j = \emptyset$  implies that  $E_i F E_j F = \emptyset$ .  $\square$

If we define  $Q(E) = P(E|F)$ , then, from Proposition 5.1,  $Q(E)$  may be regarded as a probability function on the events of  $S$ . Hence, all of the propositions previously proved for probabilities apply to  $Q(E)$ . For instance, we have

$$Q(E_1 \cup E_2) = Q(E_1) + Q(E_2) - Q(E_1 E_2)$$

or, equivalently,

$$P(E_1 \cup E_2 | F) = P(E_1 | F) + P(E_2 | F) - P(E_1 E_2 | F)$$

Also, if we define the conditional probability  $Q(E_1 | E_2)$  by  $Q(E_1 | E_2) = Q(E_1 E_2) / Q(E_2)$ , then, from Equation (3.1), we have

$$Q(E_1) = Q(E_1 | E_2) Q(E_2) + Q(E_1 | E_2^c) Q(E_2^c) \quad (5.1)$$

Since

$$\begin{aligned}
 Q(E_1 | E_2) &= \frac{Q(E_1 E_2)}{Q(E_2)} \\
 &= \frac{P(E_1 E_2 | F)}{P(E_2 | F)} \\
 &= \frac{P(E_1 E_2 F)}{P(F)} \\
 &= \frac{P(E_1 E_2 F)}{P(E_2 F)} \\
 &= P(E_1 | E_2 F)
 \end{aligned}$$

Equation (5.1) is equivalent to

$$P(E_1 | F) = P(E_1 | E_2 F) P(E_2 | F) + P(E_1 | E_2^c F) P(E_2^c | F)$$

#### Example 5a

Consider Example 3a, which is concerned with an insurance company that believes that people can be divided into two distinct classes: those who are accident prone and those who are not. During any given year, an accident-prone person will have an

accident with probability .4, whereas the corresponding figure for a person who is not prone to accidents is .2. What is the conditional probability that a new policyholder will have an accident in his or her second year of policy ownership, given that the policyholder has had an accident in the first year?

**Solution** If we let  $A$  be the event that the policyholder is accident prone and we let  $A_i, i = 1, 2$ , be the event that he or she has had an accident in the  $i$ th year, then the desired probability  $P(A_2|A_1)$  may be obtained by conditioning on whether or not the policyholder is accident prone, as follows:

$$P(A_2|A_1) = P(A_2|AA_1)P(A|A_1) + P(A_2|A^cA_1)P(A^c|A_1)$$

Now,

$$P(A|A_1) = \frac{P(A_1A)}{P(A_1)} = \frac{P(A_1|A)P(A)}{P(A)}$$

However,  $P(A)$  is assumed to equal  $\frac{3}{10}$ , and it was shown in Example 3a that  $P(A_1) = .26$ . Hence,

$$P(A|A_1) = \frac{(.4)(.3)}{.26} = \frac{6}{13}$$

Thus,

$$P(A^c|A_1) = 1 - P(A|A_1) = \frac{7}{13}$$

Since  $P(A_2|AA_1) = .4$  and  $P(A_2|A^cA_1) = .2$ , it follows that

$$P(A_2|A_1) = (.4)\frac{6}{13} + (.2)\frac{7}{13} \approx .29 \quad \blacksquare$$

#### Example 5b

A female chimp has given birth. It is not certain, however, which of two male chimps is the father. Before any genetic analysis has been performed, it is believed that the probability that male number 1 is the father is  $p$  and the probability that male number 2 is the father is  $1 - p$ . DNA obtained from the mother, male number 1, and male number 2 indicates that on one specific location of the genome, the mother has the gene pair  $(A, A)$ , male number 1 has the gene pair  $(a, a)$ , and male number 2 has the gene pair  $(A, a)$ . If a DNA test shows that the baby chimp has the gene pair  $(A, a)$ , what is the probability that male number 1 is the father?

**Solution** Let all probabilities be conditional on the event that the mother has the gene pair  $(A, A)$ , male number 1 has the gene pair  $(a, a)$ , and male number 2 has the gene pair  $(A, a)$ . Now, let  $M_i$  be the event that male number  $i, i = 1, 2$ , is the father, and let  $B_{A,a}$  be the event that the baby chimp has the gene pair  $(A, a)$ . Then,  $P(M_1|B_{A,a})$  is obtained as follows:

$$\begin{aligned} P(M_1|B_{A,a}) &= \frac{P(M_1B_{A,a})}{P(B_{A,a})} \\ &= \frac{P(B_{A,a}|M_1)P(M_1)}{P(B_{A,a}|M_1)P(M_1) + P(B_{A,a}|M_2)P(M_2)} \\ &= \frac{1 \cdot p}{1 \cdot p + (1/2)(1 - p)} \\ &= \frac{2p}{1 + p} \end{aligned}$$

Because  $\frac{2p}{1+p} > p$  when  $p < 1$ , the information that the baby's gene pair is  $(A, a)$  increases the probability that male number 1 is the father. This result is intuitive

because it is more likely that the baby would have gene pair  $(A, a)$  if  $M_1$  is true than if  $M_2$  is true (the respective conditional probabilities being 1 and  $1/2$ ). ■

The next example deals with a problem in the theory of runs.

**Example  
5c**

Independent trials, each resulting in a success with probability  $p$  or a failure with probability  $q = 1 - p$ , are performed. We are interested in computing the probability that a run of  $n$  consecutive successes occurs before a run of  $m$  consecutive failures.

**Solution** Let  $E$  be the event that a run of  $n$  consecutive successes occurs before a run of  $m$  consecutive failures. To obtain  $P(E)$ , we start by conditioning on the outcome of the first trial. That is, letting  $H$  denote the event that the first trial results in a success, we obtain

$$P(E) = pP(E|H) + qP(E|H^c) \quad (5.2)$$

Now, given that the first trial was successful, one way we can get a run of  $n$  successes before a run of  $m$  failures would be to have the next  $n - 1$  trials all result in successes. So, let us condition on whether or not that occurs. That is, letting  $F$  be the event that trials 2 through  $n$  all are successes, we obtain

$$P(E|H) = P(E|FH)P(F|H) + P(E|F^cH)P(F^c|H) \quad (5.3)$$

On the one hand, clearly,  $P(E|FH) = 1$ ; on the other hand, if the event  $F^cH$  occurs, then the first trial would result in a success, but there would be a failure some time during the next  $n - 1$  trials. However, when this failure occurs, it would wipe out all of the previous successes, and the situation would be exactly as if we started out with a failure. Hence,

$$P(E|F^cH) = P(E|H^c)$$

Because the independence of trials implies that  $F$  and  $H$  are independent, and because  $P(F) = p^{n-1}$ , it follows from Equation (5.3) that

$$P(E|H) = p^{n-1} + (1 - p^{n-1})P(E|H^c) \quad (5.4)$$

We now obtain an expression for  $P(E|H^c)$  in a similar manner. That is, we let  $G$  denote the event that trials 2 through  $m$  are all failures. Then,

$$P(E|H^c) = P(E|GH^c)P(G|H^c) + P(E|G^cH^c)P(G^c|H^c) \quad (5.5)$$

Now,  $GH^c$  is the event that the first  $m$  trials all result in failures, so  $P(E|GH^c) = 0$ . Also, if  $G^cH^c$  occurs, then the first trial is a failure, but there is at least one success in the next  $m - 1$  trials. Hence, since this success wipes out all previous failures, we see that

$$P(E|G^cH^c) = P(E|H)$$

Thus, because  $P(G^c|H^c) = P(G^c) = 1 - q^{m-1}$ , we obtain, from (5.5),

$$P(E|H^c) = (1 - q^{m-1})P(E|H) \quad (5.6)$$

Solving Equations (5.4) and (5.6) yields

$$P(E|H) = \frac{p^{n-1}}{p^{n-1} + q^{m-1} - p^{n-1}q^{m-1}}$$

and

$$P(E|H^c) = \frac{(1 - q^{m-1})p^{n-1}}{p^{n-1} + q^{m-1} - p^{n-1}q^{m-1}}$$

Thus,

$$\begin{aligned}
 P(E) &= pP(E|H) + qP(E|H^c) \\
 &= \frac{p^n + qp^{n-1}(1 - q^{m-1})}{p^{n-1} + q^{m-1} - p^{n-1}q^{m-1}} \\
 &= \frac{p^{n-1}(1 - q^m)}{p^{n-1} + q^{m-1} - p^{n-1}q^{m-1}} \quad (5.7)
 \end{aligned}$$

It is interesting to note that by the symmetry of the problem, the probability of obtaining a run of  $m$  failures before a run of  $n$  successes would be given by Equation (5.7) with  $p$  and  $q$  interchanged and  $n$  and  $m$  interchanged. Hence, this probability would equal

$$\begin{aligned}
 &P\{\text{run of } m \text{ failures before a run of } n \text{ successes}\} \\
 &= \frac{q^{m-1}(1 - p^n)}{q^{m-1} + p^{n-1} - q^{m-1}p^{n-1}} \quad (5.8)
 \end{aligned}$$

Since Equations (5.7) and (5.8) sum to 1, it follows that, with probability 1, either a run of  $n$  successes or a run of  $m$  failures will eventually occur.

As an example of Equation (5.7), we note that, in tossing a fair coin, the probability that a run of 2 heads will precede a run of 3 tails is  $\frac{7}{10}$ . For 2 consecutive heads before 4 consecutive tails, the probability rises to  $\frac{5}{6}$ . ■

In our next example, we return to a matching problem and obtain a solution by using conditional probabilities.

#### Example 5d

At a party,  $n$  men take off their hats. The hats are then mixed up, and each man randomly selects one. We say that a match occurs if a man selects his own hat. What is the probability of

- (a) no matches?
- (b) exactly  $k$  matches? ■

**Solution** (a) Let  $E$  denote the event that no matches occur, and to make explicit the dependence on  $n$ , write  $P_n = P(E)$ . We start by conditioning on whether or not the first man selects his own hat—call these events  $M$  and  $M^c$ , respectively. Then,

$$P_n = P(E) = P(E|M)P(M) + P(E|M^c)P(M^c)$$

Clearly,  $P(E|M) = 0$ , so

$$P_n = P(E|M^c) \frac{n-1}{n} \quad (5.9)$$

Now,  $P(E|M^c)$  is the probability of no matches when  $n-1$  men select from a set of  $n-1$  hats that does not contain the hat of one of these men. This can happen in either of two mutually exclusive ways: Either there are no matches and the extra man does not select the extra hat (this being the hat of the man who chose first) or there are no matches and the extra man does select the extra hat. The probability of the first of these events is just  $P_{n-1}$ , which is seen by regarding the extra hat as “belonging” to the extra man. Because the second event has probability  $[1/(n-1)]P_{n-2}$ , we have

$$P(E|M^c) = P_{n-1} + \frac{1}{n-1}P_{n-2}$$

Thus, from Equation (5.9),

$$P_n = \frac{n-1}{n}P_{n-1} + \frac{1}{n}P_{n-2}$$

or, equivalently,

$$P_n - P_{n-1} = -\frac{1}{n}(P_{n-1} - P_{n-2}) \quad (5.10)$$

However, since  $P_n$  is the probability of no matches when  $n$  men select among their own hats, we have

$$P_1 = 0 \quad P_2 = \frac{1}{2}$$

So, from Equation (5.10),

$$\begin{aligned} P_3 - P_2 &= -\frac{(P_2 - P_1)}{3} = -\frac{1}{3!} \quad \text{or} \quad P_3 = \frac{1}{2!} - \frac{1}{3!} \\ P_4 - P_3 &= -\frac{(P_3 - P_2)}{4} = \frac{1}{4!} \quad \text{or} \quad P_4 = \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \end{aligned}$$

and, in general,

$$P_n = \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \cdots + \frac{(-1)^n}{n!}$$

(b) To obtain the probability of exactly  $k$  matches, we consider any fixed group of  $k$  men. The probability that they, and only they, select their own hats is

$$\frac{1}{n} \frac{1}{n-1} \cdots \frac{1}{n-(k-1)} P_{n-k} = \frac{(n-k)!}{n!} P_{n-k}$$

where  $P_{n-k}$  is the conditional probability that the other  $n-k$  men, selecting among their own hats, have no matches. Since there are  $\binom{n}{k}$  choices of a set of  $k$  men, the desired probability of exactly  $k$  matches is

$$\frac{P_{n-k}}{k!} = \frac{\frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^{n-k}}{(n-k)!}}{k!} \quad \blacksquare$$

An important concept in probability theory is that of the conditional independence of events. We say that the events  $E_1$  and  $E_2$  are *conditionally independent* given  $F$  if given that  $F$  occurs, the conditional probability that  $E_1$  occurs is unchanged by information as to whether or not  $E_2$  occurs. More formally,  $E_1$  and  $E_2$  are said to be conditionally independent given  $F$  if

$$P(E_1|E_2F) = P(E_1|F) \quad (5.11)$$

or, equivalently,

$$P(E_1E_2|F) = P(E_1|F)P(E_2|F) \quad (5.12)$$

The notion of conditional independence can easily be extended to more than two events, and this extension is left as an exercise.

The reader should note that the concept of conditional independence was implicitly employed in Example 5a, where it was assumed that the events that a policyholder had an accident in his or her  $i$ th year,  $i = 1, 2, \dots$ , were conditionally independent given whether or not the person was accident prone. The following

example, sometimes referred to as *Laplace's rule of succession*, further illustrates the concept of conditional independence.

**Example  
5e**

**Laplace's rule of succession**

There are  $k + 1$  coins in a box. When flipped, the  $i$ th coin will turn up heads with probability  $i/k, i = 0, 1, \dots, k$ . A coin is randomly selected from the box and is then repeatedly flipped. If the first  $n$  flips all result in heads, what is the conditional probability that the  $(n + 1)$  flip will do likewise?

**Solution** Let  $C_i$  denote the event that the  $i$ th coin,  $i = 0, 1, \dots, k$ , is initially selected; let  $F_n$  denote the event that the first  $n$  flips all result in heads; and let  $H$  be the event that the  $(n + 1)$  flip is a head. The desired probability,  $P(H|F_n)$ , is now obtained as follows:

$$P(H|F_n) = \sum_{i=0}^k P(H|F_n C_i) P(C_i|F_n)$$

Now, given that the  $i$ th coin is selected, it is reasonable to assume that the outcomes will be conditionally independent, with each one resulting in a head with probability  $i/k$ . Hence,

$$P(H|F_n C_i) = P(H|C_i) = \frac{i}{k}$$

Also,

$$P(C_i|F_n) = \frac{P(C_i F_n)}{P(F_n)} = \frac{P(F_n|C_i)P(C_i)}{\sum_{j=0}^k P(F_n|C_j)P(C_j)} = \frac{(i/k)^n [1/(k + 1)]}{\sum_{j=0}^k (j/k)^n [1/(k + 1)]}$$

Thus,

$$P(H|F_n) = \frac{\sum_{i=0}^k (i/k)^{n+1}}{\sum_{j=0}^k (j/k)^n}$$

But if  $k$  is large, we can use the integral approximations

$$\begin{aligned} \frac{1}{k} \sum_{i=0}^k \left(\frac{i}{k}\right)^{n+1} &\approx \int_0^1 x^{n+1} dx = \frac{1}{n+2} \\ \frac{1}{k} \sum_{j=0}^k \left(\frac{j}{k}\right)^n &\approx \int_0^1 x^n dx = \frac{1}{n+1} \end{aligned}$$

So, for  $k$  large,

$$P(H|F_n) \approx \frac{n+1}{n+2}$$

■

**Example 5f****Updating information sequentially**

Suppose there are  $n$  mutually exclusive and exhaustive possible hypotheses, with initial (sometimes referred to as *prior*) probabilities  $P(H_i)$ ,  $\sum_{i=1}^n P(H_i) = 1$ . Now, if information that the event  $E$  has occurred is received, then the conditional probability that  $H_i$  is the true hypothesis (sometimes referred to as the *updated* or *posterior* probability of  $H_i$ ) is

$$P(H_i|E) = \frac{P(E|H_i)P(H_i)}{\sum_j P(E|H_j)P(H_j)} \quad (5.13)$$

Suppose now that we learn first that  $E_1$  has occurred and then that  $E_2$  has occurred. Then, given only the first piece of information, the conditional probability that  $H_i$  is the true hypothesis is

$$P(H_i|E_1) = \frac{P(E_1|H_i)P(H_i)}{P(E_1)} = \frac{P(E_1|H_i)P(H_i)}{\sum_j P(E_1|H_j)P(H_j)}$$

whereas given both pieces of information, the conditional probability that  $H_i$  is the true hypothesis is  $P(H_i|E_1E_2)$ , which can be computed by

$$P(H_i|E_1E_2) = \frac{P(E_1E_2|H_i)P(H_i)}{\sum_j P(E_1E_2|H_j)P(H_j)}$$

One might wonder, however, when one can compute  $P(H_i|E_1E_2)$  by using the right side of Equation (5.13) with  $E = E_2$  and with  $P(H_j)$  replaced by  $P(H_j|E_1)$ ,  $j = 1, \dots, n$ . That is, when is it legitimate to regard  $P(H_j|E_1)$ ,  $j \geq 1$ , as the prior probabilities and then use (5.13) to compute the posterior probabilities?

**Solution** The answer is that the preceding is legitimate, provided that for each  $j = 1, \dots, n$ , the events  $E_1$  and  $E_2$  are conditionally independent, given  $H_j$ . For if this is the case, then

$$P(E_1E_2|H_j) = P(E_2|H_j)P(E_1|H_j), \quad j = 1, \dots, n$$

Therefore,

$$\begin{aligned} P(H_i|E_1E_2) &= \frac{P(E_2|H_i)P(E_1|H_i)P(H_i)}{P(E_1E_2)} \\ &= \frac{P(E_2|H_i)P(E_1|H_i)}{P(E_1E_2)} \\ &= \frac{P(E_2|H_i)P(H_i|E_1)P(E_1)}{P(E_1E_2)} \\ &= \frac{P(E_2|H_i)P(H_i|E_1)}{Q(1,2)} \end{aligned}$$

where  $Q(1,2) = \frac{P(E_1E_2)}{P(E_1)}$ . Since the preceding equation is valid for all  $i$ , we obtain, upon summing,

$$1 = \sum_{i=1}^n P(H_i|E_1E_2) = \sum_{i=1}^n \frac{P(E_2|H_i)P(H_i|E_1)}{Q(1,2)}$$

showing that

$$Q(1,2) = \sum_{i=1}^n P(E_2|H_i)P(H_i|E_1)$$

and yielding the result

$$P(H_i|E_1E_2) = \frac{P(E_2|H_i)P(H_i|E_1)}{\sum_{i=1}^n P(E_2|H_i)P(H_i|E_1)}$$

For instance, suppose that one of two coins is chosen to be flipped. Let  $H_i$  be the event that coin  $i$ ,  $i = 1, 2$ , is chosen, and suppose that when coin  $i$  is flipped, it lands on heads with probability  $p_i$ ,  $i = 1, 2$ . Then, the preceding equations show that to sequentially update the probability that coin 1 is the one being flipped, given the results of the previous flips, all that must be saved after each new flip is the conditional probability that coin 1 is the coin being used. That is, it is not necessary to keep track of all earlier results. ■

## Summary

For events  $E$  and  $F$ , the conditional probability of  $E$  given that  $F$  has occurred is denoted by  $P(E|F)$  and is defined by

$$P(E|F) = \frac{P(EF)}{P(F)}$$

The identity

$$P(E_1E_2 \cdots E_n) = P(E_1)P(E_2|E_1) \cdots P(E_n|E_1 \cdots E_{n-1})$$

is known as the *multiplication rule* of probability.

A valuable identity is

$$P(E) = P(E|F)P(F) + P(E|F^c)P(F^c)$$

which can be used to compute  $P(E)$  by “conditioning” on whether  $F$  occurs.

$P(H)/P(H^c)$  is called the *odds* of the event  $H$ . The identity

$$\frac{P(H|E)}{P(H^c|E)} = \frac{P(H)P(E|H)}{P(H^c)P(E|H^c)}$$

shows that when new evidence  $E$  is obtained, the value of the odds of  $H$  becomes its old value multiplied by the ratio of the conditional probability of the new evidence when  $H$  is true to the conditional probability when  $H$  is not true.

Let  $F_i$ ,  $i = 1, \dots, n$ , be mutually exclusive events whose union is the entire sample space. The identity

$$P(F_j|E) = \frac{P(E|F_j)P(F_j)}{\sum_{i=1}^n P(E|F_i)P(F_i)}$$

is known as *Bayes's formula*. If the events  $F_i$ ,  $i = 1, \dots, n$ , are competing hypotheses, then Bayes's formula shows how to compute the conditional probabilities of these hypotheses when additional evidence  $E$  becomes available.

The denominator of Bayes's formula uses that

$$P(E) = \sum_{i=1}^n P(E|F_i)P(F_i)$$

which is called the *law of total probability*.

If  $P(EF) = P(E)P(F)$ , then we say that the events  $E$  and  $F$  are *independent*. This condition is equivalent to  $P(E|F) = P(E)$  and to  $P(F|E) = P(F)$ . Thus, the events  $E$  and  $F$  are independent if knowledge of the occurrence of one of them does not affect the probability of the other.

The events  $E_1, \dots, E_n$  are said to be independent if, for any subset  $E_{i_1}, \dots, E_{i_r}$  of them,

$$P(E_{i_1} \cdots E_{i_r}) = P(E_{i_1}) \cdots P(E_{i_r})$$

For a fixed event  $F$ ,  $P(E|F)$  can be considered to be a probability function on the events  $E$  of the sample space.

## Problems

1. Two fair dice are rolled. What is the conditional probability that at least one lands on 6 given that the dice land on different numbers?

2. If two fair dice are rolled, what is the conditional probability that the first one lands on 6 given that the sum of the dice is  $i$ ? Compute for all values of  $i$  between 2 and 12.

3. Use Equation (2.1) to compute in a hand of bridge the conditional probability that East has 3 spades given that North and South have a combined total of 8 spades.

4. What is the probability that at least one of a pair of fair dice lands on 6, given that the sum of the dice is  $i$ ,  $i = 2, 3, \dots, 12$ ?

5. An urn contains 6 white and 9 black balls. If 4 balls are to be randomly selected without replacement, what is the probability that the first 2 selected are white and the last 2 black?
6. Consider an urn containing 12 balls, of which 8 are white. A sample of size 4 is to be drawn with replacement (without replacement). What is the conditional probability (in each case) that the first and third balls drawn will be white given that the sample drawn contains exactly 3 white balls?
7. The king comes from a family of 2 children. What is the probability that the other child is his sister?
8. A couple has 2 children. What is the probability that both are girls if the older of the two is a girl?
9. Consider 3 urns. Urn *A* contains 2 white and 4 red balls, urn *B* contains 8 white and 4 red balls, and urn *C* contains 1 white and 3 red balls. If 1 ball is selected from each urn, what is the probability that the ball chosen from urn *A* was white given that exactly 2 white balls were selected?
10. Three cards are randomly selected, without replacement, from an ordinary deck of 52 playing cards. Compute the conditional probability that the first card selected is a spade given that the second and third cards are spades.
11. Two cards are randomly chosen without replacement from an ordinary deck of 52 cards. Let *B* be the event that both cards are aces, let  $A_s$  be the event that the ace of spades is chosen, and let *A* be the event that at least one ace is chosen. Find
  - (a)  $P(B|A_s)$
  - (b)  $P(B|A)$
12. A recent college graduate is planning to take the first three actuarial examinations in the coming summer. She will take the first actuarial exam in June. If she passes that exam, then she will take the second exam in July, and if she also passes that one, then she will take the third exam in September. If she fails an exam, then she is not allowed to take any others. The probability that she passes the first exam is .9. If she passes the first exam, then the conditional probability that she passes the second one is .8, and if she passes both the first and the second exams, then the conditional probability that she passes the third exam is .7.
  - (a) What is the probability that she passes all three exams?
  - (b) Given that she did not pass all three exams, what is the conditional probability that she failed the second exam?
13. Suppose that an ordinary deck of 52 cards (which contains 4 aces) is randomly divided into 4 hands of 13 cards each. We are interested in determining *p*, the probability that each hand has an ace. Let  $E_i$  be the event that the *i*th hand has exactly one ace. Determine  $p = P(E_1E_2E_3E_4)$  by using the multiplication rule.
14. An urn initially contains 5 white and 7 black balls. Each time a ball is selected, its color is noted and it is replaced in the urn along with 2 other balls of the same color. Compute the probability that
  - (a) the first 2 balls selected are black and the next 2 are white;
  - (b) of the first 4 balls selected, exactly 2 are black.
15. An ectopic pregnancy is twice as likely to develop when the pregnant woman is a smoker as it is when she is a nonsmoker. If 32 percent of women of childbearing age are smokers, what percentage of women having ectopic pregnancies are smokers?
16. Ninety-eight percent of all babies survive delivery. However, 15 percent of all births involve Cesarean (C) sections, and when a C section is performed, the baby survives 96 percent of the time. If a randomly chosen pregnant woman does not have a C section, what is the probability that her baby survives?
17. In a certain community, 36 percent of the families own a dog and 22 percent of the families that own a dog also own a cat. In addition, 30 percent of the families own a cat. What is
  - (a) the probability that a randomly selected family owns both a dog and a cat?
  - (b) the conditional probability that a randomly selected family owns a dog given that it owns a cat?
18. A total of 46 percent of the voters in a certain city classify themselves as Independents, whereas 30 percent classify themselves as Liberals and 24 percent say that they are Conservatives. In a recent local election, 35 percent of the Independents, 62 percent of the Liberals, and 58 percent of the Conservatives voted. A voter is chosen at random. Given that this person voted in the local election, what is the probability that he or she is
  - (a) an Independent?
  - (b) a Liberal?
  - (c) a Conservative?
  - (d) What percent of voters participated in the local election?
19. A total of 48 percent of the women and 37 percent of the men who took a certain “quit smoking” class remained nonsmokers for at least one year after completing the class. These people then attended a success party at the end of a year. If 62 percent of the original class was male,
  - (a) what percentage of those attending the party were women?
  - (b) what percentage of the original class attended the party?
20. Fifty-two percent of the students at a certain college are females. Five percent of the students in this college are majoring in computer science. Two percent of the students

are women majoring in computer science. If a student is selected at random, find the conditional probability that

(a) the student is female given that the student is majoring in computer science;

(b) this student is majoring in computer science given that the student is female.

**21.** A total of 500 married working couples were polled about their annual salaries, with the following information resulting:

| Wife               | Husband            |                    |
|--------------------|--------------------|--------------------|
|                    | Less than \$25,000 | More than \$25,000 |
| Less than \$25,000 | 212                | 198                |
| More than \$25,000 | 36                 | 54                 |

For instance, in 36 of the couples, the wife earned more and the husband earned less than \$25,000. If one of the couples is randomly chosen, what is

(a) the probability that the husband earns less than \$25,000?

(b) the conditional probability that the wife earns more than \$25,000 given that the husband earns more than this amount?

(c) the conditional probability that the wife earns more than \$25,000 given that the husband earns less than this amount?

**22.** A red die, a blue die, and a yellow die (all six sided) are rolled. We are interested in the probability that the number appearing on the blue die is less than that appearing on the yellow die, which is less than that appearing on the red die. That is, with  $B$ ,  $Y$ , and  $R$  denoting, respectively, the number appearing on the blue, yellow, and red die, we are interested in  $P(B < Y < R)$ .

(a) What is the probability that no two of the dice land on the same number?

(b) Given that no two of the dice land on the same number, what is the conditional probability that  $B < Y < R$ ?

(c) What is  $P(B < Y < R)$ ?

**23.** Urn I contains 2 white and 4 red balls, whereas urn II contains 1 white and 1 red ball. A ball is randomly chosen from urn I and put into urn II, and a ball is then randomly selected from urn II. What is

(a) the probability that the ball selected from urn II is white?

(b) the conditional probability that the transferred ball was white given that a white ball is selected from urn II?

**24.** Each of 2 balls is painted either black or gold and then placed in an urn. Suppose that each ball is colored black with probability  $\frac{1}{2}$  and that these events are independent.

(a) Suppose that you obtain information that the gold paint has been used (and thus at least one of the balls is painted gold). Compute the conditional probability that both balls are painted gold.

(b) Suppose now that the urn tips over and 1 ball falls out. It is painted gold. What is the probability that both balls are gold in this case? Explain.

**25.** The following method was proposed to estimate the number of people over the age of 50 who reside in a town of known population 100,000: "As you walk along the streets, keep a running count of the percentage of people you encounter who are over 50. Do this for a few days; then multiply the percentage you obtain by 100,000 to obtain the estimate." Comment on this method.

*Hint:* Let  $p$  denote the proportion of people in the town who are over 50. Furthermore, let  $\alpha_1$  denote the proportion of time that a person under the age of 50 spends in the streets, and let  $\alpha_2$  be the corresponding value for those over 50. What quantity does the method suggested estimate? When is the estimate approximately equal to  $p$ ?

**26.** Suppose that 5 percent of men and 0.25 percent of women are color blind. A color-blind person is chosen at random. What is the probability of this person being male? Assume that there are an equal number of males and females. What if the population consisted of twice as many males as females?

**27.** All the workers at a certain company drive to work and park in the company's lot. The company is interested in estimating the average number of workers in a car. Which of the following methods will enable the company to estimate this quantity? Explain your answer.

1. Randomly choose  $n$  workers, find out how many were in the cars in which they were driven, and take the average of the  $n$  values.
2. Randomly choose  $n$  cars in the lot, find out how many were driven in those cars, and take the average of the  $n$  values.

**28.** Suppose that an ordinary deck of 52 cards is shuffled and the cards are then turned over one at a time until the first ace appears. Given that the first ace is the 20th card to appear, what is the conditional probability that the card following it is the

(a) ace of spades?

(b) two of clubs?

**29.** There are 15 tennis balls in a box, of which 9 have not previously been used. Three of the balls are randomly chosen, played with, and then returned to the box. Later, another 3 balls are randomly chosen from the box. Find the probability that none of these balls has ever been used.

**30.** Consider two boxes, one containing 1 black and 1 white marble, the other 2 black and 1 white marble. A

box is selected at random, and a marble is drawn from it at random. What is the probability that the marble is black? What is the probability that the first box was the one selected given that the marble is white?

**31.** Ms. Aquina has just had a biopsy on a possibly cancerous tumor. Not wanting to spoil a weekend family event, she does not want to hear any bad news in the next few days. But if she tells the doctor to call only if the news is good, then if the doctor does not call, Ms. Aquina can conclude that the news is bad. So, being a student of probability, Ms. Aquina instructs the doctor to flip a coin. If it comes up heads, the doctor is to call if the news is good and not call if the news is bad. If the coin comes up tails, the doctor is not to call. In this way, even if the doctor doesn't call, the news is not necessarily bad. Let  $\alpha$  be the probability that the tumor is cancerous; let  $\beta$  be the conditional probability that the tumor is cancerous given that the doctor does not call.

- (a) Which should be larger,  $\alpha$  or  $\beta$ ?
- (b) Find  $\beta$  in terms of  $\alpha$ , and prove your answer in part (a).

**32.** A family has  $j$  children with probability  $p_j$ , where  $p_1 = .1, p_2 = .25, p_3 = .35, p_4 = .3$ . A child from this family is randomly chosen. Given that this child is the eldest child in the family, find the conditional probability that the family has

- (a) only 1 child;
- (b) 4 children.

Redo (a) and (b) when the randomly selected child is the youngest child of the family.

**33.** On rainy days, Joe is late to work with probability .3; on nonrainy days, he is late with probability .1. With probability .7, it will rain tomorrow.

- (a) Find the probability that Joe is early tomorrow.
- (b) Given that Joe was early, what is the conditional probability that it rained?

**34.** In Example 3f, suppose that the new evidence is subject to different possible interpretations and in fact shows only that it is 90 percent likely that the criminal possesses the characteristic in question. In this case, how likely would it be that the suspect is guilty (assuming, as before, that he has the characteristic)?

**35.** With probability .6, the present was hidden by mom; with probability .4, it was hidden by dad. When mom hides the present, she hides it upstairs 70 percent of the time and downstairs 30 percent of the time. Dad is equally likely to hide it upstairs or downstairs.

- (a) What is the probability that the present is upstairs?
- (b) Given that it is downstairs, what is the probability it was hidden by dad?

**36.** Stores  $A$ ,  $B$ , and  $C$  have 50, 75, and 100 employees, respectively, and 50, 60, and 70 percent of them

respectively are women. Resignations are equally likely among all employees, regardless of sex. One woman employee resigns. What is the probability that she works in store  $C$ ?

**37. (a)** A gambler has a fair coin and a two-headed coin in his pocket. He selects one of the coins at random; when he flips it, it shows heads. What is the probability that it is the fair coin?

**(b)** Suppose that he flips the same coin a second time and, again, it shows heads. Now what is the probability that it is the fair coin?

**(c)** Suppose that he flips the same coin a third time and it shows tails. Now what is the probability that it is the fair coin?

**38.** Urn  $A$  has 5 white and 7 black balls. Urn  $B$  has 3 white and 12 black balls. We flip a fair coin. If the outcome is heads, then a ball from urn  $A$  is selected, whereas if the outcome is tails, then a ball from urn  $B$  is selected. Suppose that a white ball is selected. What is the probability that the coin landed tails?

**39.** In Example 3a, what is the probability that someone has an accident in the second year given that he or she had no accidents in the first year?

**40.** Consider a sample of size 3 drawn in the following manner: We start with an urn containing 5 white and 7 red balls. At each stage, a ball is drawn and its color is noted. The ball is then returned to the urn, along with an additional ball of the same color. Find the probability that the sample will contain exactly

- (a) 0 white balls;
- (b) 1 white ball;
- (c) 3 white balls;
- (d) 2 white balls.

**41.** A deck of cards is shuffled and then divided into two halves of 26 cards each. A card is drawn from one of the halves; it turns out to be an ace. The ace is then placed in the second half-deck. The half is then shuffled, and a card is drawn from it. Compute the probability that this drawn card is an ace.

*Hint:* Condition on whether or not the interchanged card is selected.

**42.** Twelve percent of all U.S. households are in California. A total of 1.3 percent of all U.S. households earn more than \$250,000 per year, while a total of 3.3 percent of all California households earn more than \$250,000 per year.

- (a) What proportion of all non-California households earn more than \$250,000 per year?
- (b) Given that a randomly chosen U.S. household earns more than \$250,000 per year, what is the probability it is a California household?

**43.** There are 3 coins in a box. One is a two-headed coin, another is a fair coin, and the third is a biased coin that comes up heads 75 percent of the time. When one of the 3 coins is selected at random and flipped, it shows heads. What is the probability that it was the two-headed coin?

**44.** Three prisoners are informed by their jailer that one of them has been chosen at random to be executed and the other two are to be freed. Prisoner *A* asks the jailer to tell him privately which of his fellow prisoners will be set free, claiming that there would be no harm in divulging this information because he already knows that at least one of the two will go free. The jailer refuses to answer the question, pointing out that if *A* knew which of his fellow prisoners were to be set free, then his own probability of being executed would rise from  $\frac{1}{3}$  to  $\frac{1}{2}$  because he would then be one of two prisoners. What do you think of the jailer's reasoning?

**45.** Suppose we have 10 coins such that if the  $i$ th coin is flipped, heads will appear with probability  $i/10$ ,  $i = 1, 2, \dots, 10$ . When one of the coins is randomly selected and flipped, it shows heads. What is the conditional probability that it was the fifth coin?

**46.** In any given year, a male automobile policyholder will make a claim with probability  $p_m$  and a female policyholder will make a claim with probability  $p_f$ , where  $p_f \neq p_m$ . The fraction of the policyholders that are male is  $\alpha$ ,  $0 < \alpha < 1$ . A policyholder is randomly chosen. If  $A_i$  denotes the event that this policyholder will make a claim in year  $i$ , show that

$$P(A_2|A_1) > P(A_1)$$

Give an intuitive explanation of why the preceding inequality is true.

**47.** An urn contains 5 white and 10 black balls. A fair die is rolled and that number of balls is randomly chosen from the urn. What is the probability that all of the balls selected are white? What is the conditional probability that the die landed on 3 if all the balls selected are white?

**48.** Each of 2 cabinets identical in appearance has 2 drawers. Cabinet *A* contains a silver coin in each drawer, and cabinet *B* contains a silver coin in one of its drawers and a gold coin in the other. A cabinet is randomly selected, one of its drawers is opened, and a silver coin is found. What is the probability that there is a silver coin in the other drawer?

**49.** Prostate cancer is the most common type of cancer found in males. As an indicator of whether a male has prostate cancer, doctors often perform a test that measures the level of the prostate-specific antigen (PSA) that is produced only by the prostate gland. Although PSA levels are indicative of cancer, the test is notoriously unreliable. Indeed, the probability that a noncancerous man will have an elevated PSA level is approximately .135, increasing to

approximately .268 if the man does have cancer. If, on the basis of other factors, a physician is 70 percent certain that a male has prostate cancer, what is the conditional probability that he has the cancer given that

(a) the test indicated an elevated PSA level?

(b) the test did not indicate an elevated PSA level?

Repeat the preceding calculation, this time assuming that the physician initially believes that there is a 30 percent chance that the man has prostate cancer.

**50.** Suppose that an insurance company classifies people into one of three classes: good risks, average risks, and bad risks. The company's records indicate that the probabilities that good-, average-, and bad-risk persons will be involved in an accident over a 1-year span are, respectively, .05, .15, and .30. If 20 percent of the population is a good risk, 50 percent an average risk, and 30 percent a bad risk, what proportion of people have accidents in a fixed year? If policyholder *A* had no accidents in 2012, what is the probability that he or she is a good risk? is an average risk?

**51.** A worker has asked her supervisor for a letter of recommendation for a new job. She estimates that there is an 80 percent chance that she will get the job if she receives a strong recommendation, a 40 percent chance if she receives a moderately good recommendation, and a 10 percent chance if she receives a weak recommendation. She further estimates that the probabilities that the recommendation will be strong, moderate, and weak are .7, .2, and .1, respectively.

(a) How certain is she that she will receive the new job offer?

(b) Given that she does receive the offer, how likely should she feel that she received a strong recommendation? a moderate recommendation? a weak recommendation?

(c) Given that she does not receive the job offer, how likely should she feel that she received a strong recommendation? a moderate recommendation? a weak recommendation?

**52.** A high school student is anxiously waiting to receive mail telling her whether she has been accepted to a certain college. She estimates that the conditional probabilities of receiving notification on each day of next week, given that she is accepted and that she is rejected, are as follows:

| Day       | $P(\text{mail} \text{accepted})$ | $P(\text{mail} \text{rejected})$ |
|-----------|----------------------------------|----------------------------------|
| Monday    | .15                              | .05                              |
| Tuesday   | .20                              | .10                              |
| Wednesday | .25                              | .10                              |
| Thursday  | .15                              | .15                              |
| Friday    | .10                              | .20                              |

She estimates that her probability of being accepted is .6.

- (a) What is the probability that she receives mail on Monday?
- (b) What is the conditional probability that she receives mail on Tuesday given that she does not receive mail on Monday?
- (c) If there is no mail through Wednesday, what is the conditional probability that she will be accepted?
- (d) What is the conditional probability that she will be accepted if mail comes on Thursday?
- (e) What is the conditional probability that she will be accepted if no mail arrives that week?

**53.** A parallel system functions whenever at least one of its components works. Consider a parallel system of  $n$  components, and suppose that each component works independently with probability  $\frac{1}{2}$ . Find the conditional probability that component 1 works given that the system is functioning.

**54.** If you had to construct a mathematical model for events  $E$  and  $F$ , as described in parts (a) through (e), would you assume that they were independent events? Explain your reasoning.

- (a)  $E$  is the event that a businesswoman has blue eyes, and  $F$  is the event that her secretary has blue eyes.
- (b)  $E$  is the event that a professor owns a car, and  $F$  is the event that he is listed in the telephone book.
- (c)  $E$  is the event that a man is under 6 feet tall, and  $F$  is the event that he weighs more than 200 pounds.
- (d)  $E$  is the event that a woman lives in the United States, and  $F$  is the event that she lives in the Western Hemisphere.
- (e)  $E$  is the event that it will rain tomorrow, and  $F$  is the event that it will rain the day after tomorrow.

**55.** In a class, there are 4 first-year boys, 6 first-year girls, and 6 sophomore boys. How many sophomore girls must be present if sex and class are to be independent when a student is selected at random?

**56.** Suppose that you continually collect coupons and that there are  $m$  different types. Suppose also that each time a new coupon is obtained, it is a type  $i$  coupon with probability  $p_i, i = 1, \dots, m$ . Suppose that you have just collected your  $n$ th coupon. What is the probability that it is a new type?

*Hint:* Condition on the type of this coupon.

**57.** A simplified model for the movement of the price of a stock supposes that on each day the stock's price either moves up 1 unit with probability  $p$  or moves down 1 unit with probability  $1 - p$ . The changes on different days are assumed to be independent.

- (a) What is the probability that after 2 days the stock will be at its original price?

- (b) What is the probability that after 3 days the stock's price will have increased by 1 unit?

(c) Given that after 3 days the stock's price has increased by 1 unit, what is the probability that it went up on the first day?

**58.** Suppose that we want to generate the outcome of the flip of a fair coin, but that all we have at our disposal is a biased coin that lands on heads with some unknown probability  $p$  that need not be equal to  $\frac{1}{2}$ . Consider the following procedure for accomplishing our task:

1. Flip the coin.
2. Flip the coin again.
3. If both flips land on heads or both land on tails, return to step 1.
4. Let the result of the last flip be the result of the experiment.

- (a) Show that the result is equally likely to be either heads or tails.
- (b) Could we use a simpler procedure that continues to flip the coin until the last two flips are different and then lets the result be the outcome of the final flip?

**59.** Independent flips of a coin that lands on heads with probability  $p$  are made. What is the probability that the first four outcomes are

- (a)  $H, H, H, H$ ?
- (b)  $T, H, H, H$ ?

(c) What is the probability that the pattern  $T, H, H, H$  occurs before the pattern  $H, H, H, H$ ?

*Hint for part (c):* How can the pattern  $H, H, H, H$  occur first?

**60.** The color of a person's eyes is determined by a single pair of genes. If they are both blue-eyed genes, then the person will have blue eyes; if they are both brown-eyed genes, then the person will have brown eyes; and if one of them is a blue-eyed gene and the other a brown-eyed gene, then the person will have brown eyes. (Because of the latter fact, we say that the brown-eyed gene is *dominant* over the blue-eyed one.) A newborn child independently receives one eye gene from each of its parents, and the gene it receives from a parent is equally likely to be either of the two eye genes of that parent. Suppose that Smith and both of his parents have brown eyes, but Smith's sister has blue eyes.

- (a) What is the probability that Smith possesses a blue-eyed gene?
- (b) Suppose that Smith's wife has blue eyes. What is the probability that their first child will have blue eyes?
- (c) If their first child has brown eyes, what is the probability that their next child will also have brown eyes?

**61.** Genes relating to albinism are denoted by  $A$  and  $a$ . Only those people who receive the  $a$  gene from both

parents will be albino. Persons having the gene pair  $A, a$  are normal in appearance and, because they can pass on the trait to their offspring, are called carriers. Suppose that a normal couple has two children, exactly one of whom is an albino. Suppose that the nonalbino child mates with a person who is known to be a carrier for albinism.

(a) What is the probability that their first offspring is an albino?

(b) What is the conditional probability that their second offspring is an albino given that their firstborn is not?

**62.** Barbara and Dianne go target shooting. Suppose that each of Barbara's shots hits a wooden duck target with probability  $p_1$ , while each shot of Dianne's hits it with probability  $p_2$ . Suppose that they shoot simultaneously at the same target. If the wooden duck is knocked over (indicating that it was hit), what is the probability that

(a) both shots hit the duck?

(b) Barbara's shot hit the duck?

What independence assumptions have you made?

**63.**  $A$  and  $B$  are involved in a duel. The rules of the duel are that they are to pick up their guns and shoot at each other simultaneously. If one or both are hit, then the duel is over. If both shots miss, then they repeat the process. Suppose that the results of the shots are independent and that each shot of  $A$  will hit  $B$  with probability  $p_A$ , and each shot of  $B$  will hit  $A$  with probability  $p_B$ . What is

(a) the probability that  $A$  is not hit?

(b) the probability that both duelists are hit?

(c) the probability that the duel ends after the  $n$ th round of shots?

(d) the conditional probability that the duel ends after the  $n$ th round of shots given that  $A$  is not hit?

(e) the conditional probability that the duel ends after the  $n$ th round of shots given that both duelists are hit?

**64.** A true-false question is to be posed to a husband-and-wife team on a quiz show. Both the husband and the wife will independently give the correct answer with probability  $p$ . Which of the following is a better strategy for the couple?

(a) Choose one of them and let that person answer the question.

(b) Have them both consider the question, and then either give the common answer if they agree or, if they disagree, flip a coin to determine which answer to give.

**65.** Assume, as in Example 3h, that 64 percent of twins are of the same sex. Given that a newborn set of twins is of the same sex, what is the conditional probability that the twins are identical?

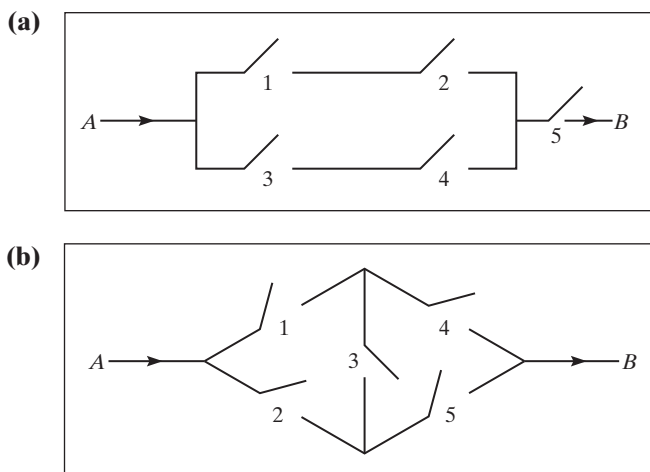
**66.** The probability of the closing of the  $i$ th relay in the circuits shown in Figure 4 is given by  $p_i, i = 1, 2, 3, 4, 5$ . If all relays function independently, what is the probability that a current flows between  $A$  and  $B$  for the respective circuits?

*Hint for (b):* Condition on whether relay 3 closes.

**67.** An engineering system consisting of  $n$  components is said to be a  $k$ -out-of- $n$  system ( $k \leq n$ ) if the system functions if and only if at least  $k$  of the  $n$  components function. Suppose that all components function independently of one another.

(a) If the  $i$ th component functions with probability  $P_i, i = 1, 2, 3, 4$ , compute the probability that a 2-out-of-4 system functions.

(b) Repeat part (a) for a 3-out-of-5 system.



**Figure 4** Circuits for Problem 66.

(c) Repeat for a  $k$ -out-of- $n$  system when all the  $P_i$  equal  $p$  (that is,  $P_i = p, i = 1, 2, \dots, n$ ).

**68.** In Problem 66a, find the conditional probability that relays 1 and 2 are both closed given that a current flows from  $A$  to  $B$ .

**69.** A certain organism possesses a pair of each of 5 different genes (which we will designate by the first 5 letters of the English alphabet). Each gene appears in 2 forms (which we designate by lowercase and capital letters). The capital letter will be assumed to be the dominant gene, in the sense that if an organism possesses the gene pair  $xX$ , then it will outwardly have the appearance of the  $X$  gene. For instance, if  $X$  stands for brown eyes and  $x$  for blue eyes, then an individual having either gene pair  $XX$  or  $xX$  will have brown eyes, whereas one having gene pair  $xx$  will have blue eyes. The characteristic appearance of an organism is called its phenotype, whereas its genetic constitution is called its genotype. (Thus, 2 organisms with respective genotypes  $aA, bB, cC, dD, eE$  and  $AA, BB, CC, DD, EE$  would have different genotypes but the same phenotype.) In a mating between 2 organisms, each one contributes, at random, one of its gene pairs of each type. The 5 contributions of an organism (one of each of the 5 types) are assumed to be independent and are also independent of the contributions of the organism's mate. In a mating between organisms having genotypes  $aA, bB, cC, dD, eE$  and  $aa, bb, cc, dd, ee$  what is the probability that the progeny will (i) phenotypically and (ii) genotypically resemble

- (a) the first parent?
- (b) the second parent?
- (c) either parent?
- (d) neither parent?

**70.** There is a 50–50 chance that the queen carries the gene for hemophilia. If she is a carrier, then each prince has a 50–50 chance of having hemophilia. If the queen has had three princes without the disease, what is the probability that the queen is a carrier? If there is a fourth prince, what is the probability that he will have hemophilia?

**71.** On the morning of September 30, 1982, the won–lost records of the three leading baseball teams in the Western Division of the National League were as follows:

| Team                 | Won | Lost |
|----------------------|-----|------|
| Atlanta Braves       | 87  | 72   |
| San Francisco Giants | 86  | 73   |
| Los Angeles Dodgers  | 86  | 73   |

Each team had 3 games remaining. All 3 of the Giants' games were with the Dodgers, and the 3 remaining games of the Braves were against the San Diego Padres. Suppose that the outcomes of all remaining games are independent

and each game is equally likely to be won by either participant. For each team, what is the probability that it will win the division title? If two teams tie for first place, they have a playoff game, which each team has an equal chance of winning.

**72.** A town council of 7 members contains a steering committee of size 3. New ideas for legislation go first to the steering committee and then on to the council as a whole if at least 2 of the 3 committee members approve the legislation. Once at the full council, the legislation requires a majority vote (of at least 4) to pass. Consider a new piece of legislation, and suppose that each town council member will approve it, independently, with probability  $p$ . What is the probability that a given steering committee member's vote is decisive in the sense that if that person's vote were reversed, then the final fate of the legislation would be reversed? What is the corresponding probability for a given council member not on the steering committee?

**73.** Suppose that each child born to a couple is equally likely to be a boy or a girl, independently of the sex distribution of the other children in the family. For a couple having 5 children, compute the probabilities of the following events:

- (a) All children are of the same sex.
- (b) The 3 eldest are boys and the others girls.
- (c) Exactly 3 are boys.
- (d) The 2 oldest are girls.
- (e) There is at least 1 girl.

**74.**  $A$  and  $B$  alternate rolling a pair of dice, stopping either when  $A$  rolls the sum 9 or when  $B$  rolls the sum 6. Assuming that  $A$  rolls first, find the probability that the final roll is made by  $A$ .

**75.** In a certain village, it is traditional for the eldest son (or the older son in a two-son family) and his wife to be responsible for taking care of his parents as they age. In recent years, however, the women of this village, not wanting that responsibility, have not looked favorably upon marrying an eldest son.

- (a) If every family in the village has two children, what proportion of all sons are older sons?
  - (b) If every family in the village has three children, what proportion of all sons are eldest sons?
- Assume that each child is, independently, equally likely to be either a boy or a girl.

**76.** Suppose that  $E$  and  $F$  are mutually exclusive events of an experiment. Show that if independent trials of this experiment are performed, then  $E$  will occur before  $F$  with probability  $P(E)/[P(E) + P(F)]$ .

**77.** Consider an unending sequence of independent trials, where each trial is equally likely to result in any of the outcomes 1, 2, or 3. Given that outcome 3 is the last of the

three outcomes to occur, find the conditional probability that

- (a) the first trial results in outcome 1;
- (b) the first two trials both result in outcome 1.

**78.** *A* and *B* play a series of games. Each game is independently won by *A* with probability  $p$  and by *B* with probability  $1 - p$ . They stop when the total number of wins of one of the players is two greater than that of the other player. The player with the greater number of total wins is declared the winner of the series.

- (a) Find the probability that a total of 4 games are played.
- (b) Find the probability that *A* is the winner of the series.

**79.** In successive rolls of a pair of fair dice, what is the probability of getting 2 sevens before 6 even numbers?

**80.** In a certain contest, the players are of equal skill and the probability is  $\frac{1}{2}$  that a specified one of the two contestants will be the victor. In a group of  $2^n$  players, the players are paired off against each other at random. The  $2^{n-1}$  winners are again paired off randomly, and so on, until a single winner remains. Consider two specified contestants, *A* and *B*, and define the events  $A_i, i \leq n, E$  by

$A_i$ : *A* plays in exactly  $i$  contests  
 $E$ : *A* and *B* never play each other

- (a) Find  $P(A_i), i = 1, \dots, n$ .
- (b) Find  $P(E)$ .
- (c) Let  $P_n = P(E)$ . Show that

$$P_n = \frac{1}{2^n - 1} + \frac{2^n - 2}{2^n - 1} \left(\frac{1}{2}\right)^2 P_{n-1}$$

and use this formula to check the answer you obtained in part (b).

*Hint:* Find  $P(E)$  by conditioning on which of the events  $A_i, i = 1, \dots, n$  occur. In simplifying your answer, use the algebraic identity

$$\sum_{i=1}^{n-1} ix^{i-1} = \frac{1 - nx^{n-1} + (n-1)x^n}{(1-x)^2}$$

For another approach to solving this problem, note that there are a total of  $2^n - 1$  games played.

- (d) Explain why  $2^n - 1$  games are played. Number these games, and let  $B_i$  denote the event that *A* and *B* play each other in game  $i, i = 1, \dots, 2^n - 1$ .
- (e) What is  $P(B_i)$ ?
- (f) Use part (e) to find  $P(E)$ .

**81.** An investor owns shares in a stock whose present value is 25. She has decided that she must sell her stock if it goes either down to 10 or up to 40. If each change of price is either up 1 point with probability .55 or down 1 point

with probability .45, and the successive changes are independent, what is the probability that the investor retires a winner?

**82.** *A* and *B* flip coins. *A* starts and continues flipping until a tail occurs, at which point *B* starts flipping and continues until there is a tail. Then *A* takes over, and so on. Let  $P_1$  be the probability of the coin landing on heads when *A* flips and  $P_2$  when *B* flips. The winner of the game is the first one to get

- (a) 2 heads in a row;
- (b) a total of 2 heads;
- (c) 3 heads in a row;
- (d) a total of 3 heads.

In each case, find the probability that *A* wins.

**83.** Die *A* has 4 red and 2 white faces, whereas die *B* has 2 red and 4 white faces. A fair coin is flipped once. If it lands on heads, the game continues with die *A*; if it lands on tails, then die *B* is to be used.

- (a) Show that the probability of red at any throw is  $\frac{1}{2}$ .
- (b) If the first two throws result in red, what is the probability of red at the third throw?
- (c) If red turns up at the first two throws, what is the probability that it is die *A* that is being used?

**84.** An urn contains 12 balls, of which 4 are white. Three players—*A*, *B*, and *C*—successively draw from the urn, *A* first, then *B*, then *C*, then *A*, and so on. The winner is the first one to draw a white ball. Find the probability of winning for each player if

- (a) each ball is replaced after it is drawn;
- (b) the balls that are withdrawn are not replaced.

**85.** Repeat Problem 84 when each of the 3 players selects from his own urn. That is, suppose that there are 3 different urns of 12 balls with 4 white balls in each urn.

**86.** Let  $S = \{1, 2, \dots, n\}$  and suppose that *A* and *B* are, independently, equally likely to be any of the  $2^n$  subsets (including the null set and *S* itself) of *S*.

- (a) Show that

$$P\{A \subset B\} = \left(\frac{3}{4}\right)^n$$

*Hint:* Let  $N(B)$  denote the number of elements in *B*. Use

$$P\{A \subset B\} = \sum_{i=0}^n P\{A \subset B | N(B) = i\} P\{N(B) = i\}$$

Show that  $P\{AB = \emptyset\} = \left(\frac{3}{4}\right)^n$ .

**87.** Consider Example 2a, but now suppose that when the key is in a certain pocket, there is a 10 percent chance that a search of that pocket will not find the key. Let *R* and *L* be, respectively, the events that the key is in the right-hand pocket of the jacket and that it is in the left-hand

pocket. Also, let  $S_R$  be the event that a search of the right-hand jacket pocket will be successful in finding the key, and let  $U_L$  be the event that a search of the left-hand jacket pocket will be unsuccessful and, thus, not find the key. Find  $P(S_R|U_L)$ , the conditional probability that a search of the right-hand pocket will find the key given that a search of the left-hand pocket did not, by

(a) using the identity

$$P(S_R|U_L) = \frac{P(S_R U_L)}{P(U_L)}$$

determining  $P(S_R U_L)$  by conditioning on whether or not the key is in the right-hand pocket, and determining  $P(U_L)$  by conditioning on whether or not the key is in the left-hand pocket;

(b) using the identity

$$P(S_R|U_L) = P(S_R|R U_L)P(R|U_L) + P(S_R|R^c U_L)P(R^c|U_L)$$

88. In Example 5e, what is the conditional probability that the  $i$ th coin was selected given that the first  $n$  trials all result in heads?

89. In Laplace's rule of succession (Example 5e), are the outcomes of the successive flips independent? Explain.

90. A person tried by a 3-judge panel is declared guilty if at least 2 judges cast votes of guilty. Suppose that when the defendant is in fact guilty, each judge will independently vote guilty with probability .7, whereas when the defendant is in fact innocent, this probability drops to .2. If 70 percent of defendants are guilty, compute the conditional probability that judge number 3 votes guilty given that

(a) judges 1 and 2 vote guilty;

(b) judges 1 and 2 cast 1 guilty and 1 not guilty vote;

(c) judges 1 and 2 both cast not guilty votes.

Let  $E_i, i = 1, 2, 3$  denote the event that judge  $i$  casts a guilty vote. Are these events independent? Are they conditionally independent? Explain.

91. Suppose that  $n$  independent trials, each of which results in any of the outcomes 0, 1, or 2, with respective probabilities  $p_0, p_1$ , and  $p_2$ ,  $\sum_{i=0}^2 p_i = 1$ , are performed. Find the probability that outcomes 1 and 2 both occur at least once.

## Theoretical Exercises

1. Show that if  $P(A) > 0$ , then

$$P(AB|A) \geq P(AB|A \cup B)$$

2. Let  $A \subset B$ . Express the following probabilities as simply as possible:

$$P(A|B), \quad P(A|B^c), \quad P(B|A), \quad P(B|A^c)$$

3. Consider a school community of  $m$  families, with  $n_i$  of them having  $i$  children,  $i = 1, \dots, k$ ,  $\sum_{i=1}^k n_i = m$ . Consider the following two methods for choosing a child:

1. Choose one of the  $m$  families at random and then randomly choose a child from that family.

2. Choose one of the  $\sum_{i=1}^k i n_i$  children at random.

Show that method 1 is more likely than method 2 to result in the choice of a firstborn child.

Hint: In solving this problem, you will need to show that

$$\sum_{i=1}^k i n_i \sum_{j=1}^k \frac{n_j}{j} \geq \sum_{i=1}^k n_i \sum_{j=1}^k n_j$$

To do so, multiply the sums and show that for all pairs  $i, j$ , the coefficient of the term  $n_i n_j$  is greater in the expression on the left than in the one on the right.

4. A ball is in any one of  $n$  boxes and is in the  $i$ th box with probability  $P_i$ . If the ball is in box  $i$ , a search of that box will uncover it with probability  $\alpha_i$ . Show that the conditional probability that the ball is in box  $j$ , given that a search of box  $i$  did not uncover it, is

$$\frac{P_j}{1 - \alpha_i P_i} \quad \text{if } j \neq i$$

$$\frac{(1 - \alpha_i) P_i}{1 - \alpha_i P_i} \quad \text{if } j = i$$

5. (a) Prove that if  $E$  and  $F$  are mutually exclusive, then

$$P(E|E \cup F) = \frac{P(E)}{P(E) + P(F)}$$

(b) Prove that if  $E_i, i \geq 1$  are mutually exclusive, then

$$P(E_j | \bigcup_{i=1}^{\infty} E_i) = \frac{P(E_j)}{\sum_{i=1}^{\infty} P(E_i)}$$

6. Prove that if  $E_1, E_2, \dots, E_n$  are independent events, then

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = 1 - \prod_{i=1}^n [1 - P(E_i)]$$

7. (a) An urn contains  $n$  white and  $m$  black balls. The balls are withdrawn one at a time until only those of the same color are left. Show that with probability  $n/(n + m)$ , they are all white.

*Hint:* Imagine that the experiment continues until all the balls are removed, and consider the last ball withdrawn.

(b) A pond contains 3 distinct species of fish, which we will call the Red, Blue, and Green fish. There are  $r$  Red,  $b$  Blue, and  $g$  Green fish. Suppose that the fish are removed from the pond in a random order. (That is, each selection is equally likely to be any of the remaining fish.) What is the probability that the Red fish are the first species to become extinct in the pond?

*Hint:* Write  $P\{R\} = P\{RBG\} + P\{RGB\}$ , and compute the probabilities on the right by first conditioning on the last species to be removed.

8. Let  $A, B$ , and  $C$  be events relating to the experiment of rolling a pair of dice.

(a) If

$$P(A|C) > P(B|C) \quad \text{and} \quad P(A|C^c) > P(B|C^c)$$

either prove that  $P(A) > P(B)$  or give a counterexample by defining events  $A, B$ , and  $C$  for which that relationship is not true.

(b) If

$$P(A|C) > P(A|C^c) \quad \text{and} \quad P(B|C) > P(B|C^c)$$

either prove that  $P(AB|C) > P(AB|C^c)$  or give a counterexample by defining events  $A, B$ , and  $C$  for which that relationship is not true.

*Hint:* Let  $C$  be the event that the sum of a pair of dice is 10; let  $A$  be the event that the first die lands on 6; let  $B$  be the event that the second die lands on 6.

9. Consider two independent tosses of a fair coin. Let  $A$  be the event that the first toss results in heads, let  $B$  be the event that the second toss results in heads, and let  $C$  be the event that in both tosses the coin lands on the same side. Show that the events  $A, B$ , and  $C$  are pairwise independent—that is,  $A$  and  $B$  are independent,  $A$  and  $C$  are independent, and  $B$  and  $C$  are independent—but not independent.

10. Two percent of women age 45 who participate in routine screening have breast cancer. Ninety percent of those with breast cancer have positive mammographies. Eight percent of the women who do not have breast cancer will also have positive mammographies. Given that a woman

has a positive mammography, what is the probability she has breast cancer?

11. In each of  $n$  independent tosses of a coin, the coin lands on heads with probability  $p$ . How large need  $n$  be so that the probability of obtaining at least one head is at least  $\frac{1}{2}$ ?

12. Show that  $0 \leq a_i \leq 1, i = 1, 2, \dots$ , then

$$\sum_{i=1}^{\infty} \left[ a_i \prod_{j=1}^{i-1} (1 - a_j) \right] + \prod_{i=1}^{\infty} (1 - a_i) = 1$$

*Hint:* Suppose that an infinite number of coins are to be flipped. Let  $a_i$  be the probability that the  $i$ th coin lands on heads, and consider when the first head occurs.

13. The probability of getting a head on a single toss of a coin is  $p$ . Suppose that  $A$  starts and continues to flip the coin until a tail shows up, at which point  $B$  starts flipping. Then  $B$  continues to flip until a tail comes up, at which point  $A$  takes over, and so on. Let  $P_{n,m}$  denote the probability that  $A$  accumulates a total of  $n$  heads before  $B$  accumulates  $m$ . Show that

$$P_{n,m} = pP_{n-1,m} + (1 - p)(1 - P_{m,n})$$

\*14. Suppose that you are gambling against an infinitely rich adversary and at each stage you either win or lose 1 unit with respective probabilities  $p$  and  $1 - p$ . Show that the probability that you eventually go broke is

$$\begin{aligned} &1 \quad \text{if } p \leq \frac{1}{2} \\ &(q/p)^i \quad \text{if } p > \frac{1}{2} \end{aligned}$$

where  $q = 1 - p$  and where  $i$  is your initial fortune.

15. Independent trials that result in a success with probability  $p$  are successively performed until a total of  $r$  successes is obtained. Show that the probability that exactly  $n$  trials are required is

$$\binom{n-1}{r-1} p^r (1-p)^{n-r}$$

Use this result to solve the problem of the points (Example 4j).

*Hint:* In order for it to take  $n$  trials to obtain  $r$  successes, how many successes must occur in the first  $n - 1$  trials?

16. Independent trials that result in a success with probability  $p$  and a failure with probability  $1 - p$  are called *Bernoulli trials*. Let  $P_n$  denote the probability that  $n$  Bernoulli trials result in an even number of successes (0 being considered an even number). Show that

$$P_n = p(1 - P_{n-1}) + (1 - p)P_{n-1} \quad n \geq 1$$

and use this formula to prove (by induction) that

$$P_n = \frac{1 + (1 - 2p)^n}{2}$$

**17.** Suppose that  $n$  independent trials are performed, with trial  $i$  being a success with probability  $1/(2i + 1)$ . Let  $P_n$  denote the probability that the total number of successes that result is an odd number.

- (a) Find  $P_n$  for  $n = 1, 2, 3, 4, 5$ .
- (b) Conjecture a general formula for  $P_n$ .
- (c) Derive a formula for  $P_n$  in terms of  $P_{n-1}$ .
- (d) Verify that your conjecture in part (b) satisfies the recursive formula in part (c). Because the recursive formula has a unique solution, this then proves that your conjecture is correct.

**18.** Let  $Q_n$  denote the probability that no run of 3 consecutive heads appears in  $n$  tosses of a fair coin. Show that

$$Q_n = \frac{1}{2}Q_{n-1} + \frac{1}{4}Q_{n-2} + \frac{1}{8}Q_{n-3}$$

$$Q_0 = Q_1 = Q_2 = 1$$

Find  $Q_8$ .

*Hint:* Condition on the first tail.

**19.** Consider the gambler's ruin problem, with the exception that  $A$  and  $B$  agree to play no more than  $n$  games. Let  $P_{n,i}$  denote the probability that  $A$  winds up with all the money when  $A$  starts with  $i$  and  $B$  starts with  $N - i$ . Derive an equation for  $P_{n,i}$  in terms of  $P_{n-1, i+1}$  and  $P_{n-1, i-1}$ , and compute  $P_{7,3}$ ,  $N = 5$ .

**20.** Consider two urns, each containing both white and black balls. The probabilities of drawing white balls from the first and second urns are, respectively,  $p$  and  $p'$ . Balls are sequentially selected with replacement as follows: With probability  $\alpha$ , a ball is initially chosen from the first urn, and with probability  $1 - \alpha$ , it is chosen from the second urn. The subsequent selections are then made according to the rule that whenever a white ball is drawn (and replaced), the next ball is drawn from the same urn, but when a black ball is drawn, the next ball is taken from the other urn. Let  $\alpha_n$  denote the probability that the  $n$ th ball is chosen from the first urn. Show that

$$\alpha_{n+1} = \alpha_n(p + p' - 1) + 1 - p' \quad n \geq 1$$

and use this formula to prove that

$$\alpha_n = \frac{1 - p'}{2 - p - p'} + \left( \alpha - \frac{1 - p'}{2 - p - p'} \right) \times (p + p' - 1)^{n-1}$$

Let  $P_n$  denote the probability that the  $n$ th ball selected is white. Find  $P_n$ . Also, compute  $\lim_{n \rightarrow \infty} \alpha_n$  and  $\lim_{n \rightarrow \infty} P_n$ .

**21. The Ballot Problem.** In an election, candidate  $A$  receives  $n$  votes and candidate  $B$  receives  $m$  votes, where  $n > m$ . Assuming that all of the  $(n + m)!/n!m!$  orderings of the votes are equally likely, let  $P_{n,m}$  denote the probability that  $A$  is always ahead in the counting of the votes.

- (a) Compute  $P_{2,1}, P_{3,1}, P_{3,2}, P_{4,1}, P_{4,2}, P_{4,3}$ .
- (b) Find  $P_{n,1}, P_{n,2}$ .
- (c) On the basis of your results in parts (a) and (b), conjecture the value of  $P_{n,m}$ .
- (d) Derive a recursion for  $P_{n,m}$  in terms of  $P_{n-1,m}$  and  $P_{n,m-1}$  by conditioning on who receives the last vote.
- (e) Use part (d) to verify your conjecture in part (c) by an induction proof on  $n + m$ .

**22.** As a simplified model for weather forecasting, suppose that the weather (either wet or dry) tomorrow will be the same as the weather today with probability  $p$ . Show that the weather is dry on January 1, then  $P_n$ , the probability that it will be dry  $n$  days later, satisfies

$$P_n = (2p - 1)P_{n-1} + (1 - p) \quad n \geq 1$$

$$P_0 = 1$$

Prove that

$$P_n = \frac{1}{2} + \frac{1}{2}(2p - 1)^n \quad n \geq 0$$

**23.** A bag contains  $a$  white and  $b$  black balls. Balls are chosen from the bag according to the following method:

1. A ball is chosen at random and is discarded.
2. A second ball is then chosen. If its color is different from that of the preceding ball, it is replaced in the bag and the process is repeated from the beginning. If its color is the same, it is discarded and we start from step 2.

In other words, balls are sampled and discarded until a change in color occurs, at which point the last ball is returned to the urn and the process starts anew. Let  $P_{a,b}$  denote the probability that the last ball in the bag is white. Prove that

$$P_{a,b} = \frac{1}{2}$$

*Hint:* Use induction on  $k \equiv a + b$ .

**\*24.** A round-robin tournament of  $n$  contestants is a tournament in which each of the  $\binom{n}{2}$  pairs of contestants play each other exactly once, with the outcome of any play being that one of the contestants wins and the other loses. For a fixed integer  $k$ ,  $k < n$ , a question of interest is whether it is possible that the tournament outcome is such

that for every set of  $k$  players, there is a player who beat each member of that set. Show that if

$$\binom{n}{k} \left[ 1 - \left( \frac{1}{2} \right)^k \right]^{n-k} < 1$$

then such an outcome is possible.

*Hint:* Suppose that the results of the games are independent and that each game is equally likely to be won by either contestant. Number the  $\binom{n}{k}$  sets of  $k$  contestants, and let  $B_i$  denote the event that no contestant beat all of the  $k$  players in the  $i$ th set. Then use Boole's inequality to bound  $P\left(\bigcup_i B_i\right)$ .

**25.** Prove directly that

$$P(E|F) = P(E|FG)P(G|F) + P(E|FG^c)P(G^c|F)$$

**26.** Prove the equivalence of Equations (5.11) and (5.12).

**27.** Extend the definition of conditional independence to more than 2 events.

**28.** Prove or give a counterexample. If  $E_1$  and  $E_2$  are independent, then they are conditionally independent given  $F$ .

**29.** In Laplace's rule of succession (Example 5e), show that if the first  $n$  flips all result in heads, then the conditional probability that the next  $m$  flips also result in all heads is  $(n + 1)/(n + m + 1)$ .

**30.** In Laplace's rule of succession (Example 5e), suppose that the first  $n$  flips resulted in  $r$  heads and  $n - r$  tails. Show that the probability that the  $(n + 1)$  flip turns up heads is  $(r + 1)/(n + 2)$ . To do so, you will have to prove and use the identity

$$\int_0^1 y^n (1 - y)^m dy = \frac{n!m!}{(n + m + 1)!}$$

*Hint:* To prove the identity, let  $C(n, m) = \int_0^1 y^n (1 - y)^m dy$ . Integrating by parts yields

$$C(n, m) = \frac{m}{n + 1} C(n + 1, m - 1)$$

Starting with  $C(n, 0) = 1/(n + 1)$ , prove the identity by induction on  $m$ .

**31.** Suppose that a nonmathematical, but philosophically minded, friend of yours claims that Laplace's rule of succession must be incorrect because it can lead to ridiculous conclusions. "For instance," says he, "the rule states that if a boy is 10 years old, having lived 10 years, the boy has probability  $\frac{11}{12}$  of living another year. On the other hand, if the boy has an 80-year-old grandfather, then, by Laplace's rule, the grandfather has probability  $\frac{81}{82}$  of surviving another year. However, this is ridiculous. Clearly, the boy is more likely to survive an additional year than the grandfather is." How would you answer your friend?

## Self-Test Problems and Exercises

**1.** In a game of bridge, West has no aces. What is the probability of his partner's having (a) no aces? (b) 2 or more aces? (c) What would the probabilities be if West had exactly 1 ace?

**2.** The probability that a new car battery functions for more than 10,000 miles is .8, the probability that it functions for more than 20,000 miles is .4, and the probability that it functions for more than 30,000 miles is .1. If a new car battery is still working after 10,000 miles, what is the probability that

- (a) its total life will exceed 20,000 miles?
- (b) its additional life will exceed 20,000 miles?

**3.** How can 20 balls, 10 white and 10 black, be put into two urns so as to maximize the probability of drawing a white ball if an urn is selected at random and a ball is drawn at random from it?

**4.** Urn  $A$  contains 2 white balls and 1 black ball, whereas urn  $B$  contains 1 white ball and 5 black balls. A ball is drawn at random from urn  $A$  and placed in urn  $B$ . A

ball is then drawn from urn  $B$ . It happens to be white. What is the probability that the ball transferred was white?

**5.** An urn has  $r$  red and  $w$  white balls that are randomly removed one at a time. Let  $R_i$  be the event that the  $i$ th ball removed is red. Find

- (a)  $P(R_i)$
- (b)  $P(R_5|R_3)$
- (c)  $P(R_3|R_5)$

**6.** An urn contains  $b$  black balls and  $r$  red balls. One of the balls is drawn at random, but when it is put back in the urn,  $c$  additional balls of the same color are put in with it. Now, suppose that we draw another ball. Show that the probability that the first ball was black, given that the second ball drawn was red, is  $b/(b + r + c)$ .

**7.** A friend randomly chooses two cards, without replacement, from an ordinary deck of 52 playing cards. In each of the following situations, determine the conditional probability that both cards are aces.

- (a) You ask your friend if one of the cards is the ace of spades, and your friend answers in the affirmative.  
 (b) You ask your friend if the first card selected is an ace, and your friend answers in the affirmative.  
 (c) You ask your friend if the second card selected is an ace, and your friend answers in the affirmative.  
 (d) You ask your friend if either of the cards selected is an ace, and your friend answers in the affirmative.

8. Show that

$$\frac{P(H|E)}{P(G|E)} = \frac{P(H)}{P(G)} \frac{P(E|H)}{P(E|G)}$$

Suppose that, before new evidence is observed, the hypothesis  $H$  is three times as likely to be true as is the hypothesis  $G$ . If the new evidence is twice as likely when  $G$  is true than it is when  $H$  is true, which hypothesis is more likely after the evidence has been observed?

9. You ask your neighbor to water a sickly plant while you are on vacation. Without water, it will die with probability .8; with water, it will die with probability .15. You are 90 percent certain that your neighbor will remember to water the plant.

- (a) What is the probability that the plant will be alive when you return?  
 (b) If the plant is dead upon your return, what is the probability that your neighbor forgot to water it?

10. Six balls are to be randomly chosen from an urn containing 8 red, 10 green, and 12 blue balls.

- (a) What is the probability at least one red ball is chosen?  
 (b) Given that no red balls are chosen, what is the conditional probability that there are exactly 2 green balls among the 6 chosen?

11. A type C battery is in working condition with probability .7, whereas a type D battery is in working condition with probability .4. A battery is randomly chosen from a bin consisting of 8 type C and 6 type D batteries.

- (a) What is the probability that the battery works?  
 (b) Given that the battery does not work, what is the conditional probability that it was a type C battery?

12. Maria will take two books with her on a trip. Suppose that the probability that she will like book 1 is .6, the probability that she will like book 2 is .5, and the probability that she will like both books is .4. Find the conditional probability that she will like book 2 given that she did not like book 1.

13. Balls are randomly removed from an urn that initially contains 20 red and 10 blue balls.

- (a) What is the probability that all of the red balls are removed before all of the blue ones have been removed? Now suppose that the urn initially contains 20 red, 10 blue, and 8 green balls.

(b) Now what is the probability that all of the red balls are removed before all of the blue ones have been removed?

(c) What is the probability that the colors are depleted in the order blue, red, green?

(d) What is the probability that the group of blue balls is the first of the three groups to be removed?

14. A coin having probability .8 of landing on heads is flipped.  $A$  observes the result—either heads or tails—and rushes off to tell  $B$ . However, with probability .4,  $A$  will have forgotten the result by the time he reaches  $B$ . If  $A$  has forgotten, then, rather than admitting this to  $B$ , he is equally likely to tell  $B$  that the coin landed on heads or that it landed tails. (If he does remember, then he tells  $B$  the correct result.)

- (a) What is the probability that  $B$  is told that the coin landed on heads?  
 (b) What is the probability that  $B$  is told the correct result?  
 (c) Given that  $B$  is told that the coin landed on heads, what is the probability that it did in fact land on heads?

15. In a certain species of rats, black dominates over brown. Suppose that a black rat with two black parents has a brown sibling.

- (a) What is the probability that this rat is a pure black rat (as opposed to being a hybrid with one black and one brown gene)?  
 (b) Suppose that when the black rat is mated with a brown rat, all 5 of their offspring are black. Now what is the probability that the rat is a pure black rat?

16. (a) In Problem 66b, find the probability that a current flows from  $A$  to  $B$ , by conditioning on whether relay 1 closes.

(b) Find the conditional probability that relay 3 is closed given that a current flows from  $A$  to  $B$ .

17. For the  $k$ -out-of- $n$  system described in Problem 67, assume that each component independently works with probability  $\frac{1}{2}$ . Find the conditional probability that component 1 is working, given that the system works, when

- (a)  $k = 1, n = 2$ ;  
 (b)  $k = 2, n = 3$ .

18. Mr. Jones has devised a gambling system for winning at roulette. When he bets, he bets on red and places a bet only when the 10 previous spins of the roulette have landed on a black number. He reasons that his chance of winning is quite large because the probability of 11 consecutive spins resulting in black is quite small. What do you think of this system?

19. Three players simultaneously toss coins. The coin tossed by  $A(B)[C]$  turns up heads with probability  $P_1(P_2)[P_3]$ . If one person gets an outcome different from those of the other two, then he is the odd man out. If there is no odd man out, the players flip again and continue to do

so until they get an odd man out. What is the probability that  $A$  will be the odd man?

**20.** Suppose that there are  $n$  possible outcomes of a trial, with outcome  $i$  resulting with probability  $p_i, i = 1, \dots, n, \sum_{i=1}^n p_i = 1$ . If two independent trials are observed, what is the probability that the result of the second trial is larger than that of the first?

**21.** If  $A$  flips  $n + 1$  and  $B$  flips  $n$  fair coins, show that the probability that  $A$  gets more heads than  $B$  is  $\frac{1}{2}$ .

*Hint:* Condition on which player has more heads after each has flipped  $n$  coins. (There are three possibilities.)

**22.** Prove or give counterexamples to the following statements:

(a) If  $E$  is independent of  $F$  and  $E$  is independent of  $G$ , then  $E$  is independent of  $F \cup G$ .

(b) If  $E$  is independent of  $F$ , and  $E$  is independent of  $G$ , and  $FG = \emptyset$ , then  $E$  is independent of  $F \cup G$ .

(c) If  $E$  is independent of  $F$ , and  $F$  is independent of  $G$ , and  $E$  is independent of  $FG$ , then  $G$  is independent of  $EF$ .

**23.** Let  $A$  and  $B$  be events having positive probability. State whether each of the following statements is (i) necessarily true, (ii) necessarily false, or (iii) possibly true.

(a) If  $A$  and  $B$  are mutually exclusive, then they are independent.

(b) If  $A$  and  $B$  are independent, then they are mutually exclusive.

(c)  $P(A) = P(B) = .6$ , and  $A$  and  $B$  are mutually exclusive.

(d)  $P(A) = P(B) = .6$ , and  $A$  and  $B$  are independent.

**24.** Rank the following from most likely to least likely to occur:

1. A fair coin lands on heads.
2. Three independent trials, each of which is a success with probability .8, all result in successes.
3. Seven independent trials, each of which is a success with probability .9, all result in successes.

**25.** Two local factories,  $A$  and  $B$ , produce radios. Each radio produced at factory  $A$  is defective with probability .05, whereas each one produced at factory  $B$  is defective with probability .01. Suppose you purchase two radios that were produced at the same factory, which is equally likely to have been either factory  $A$  or factory  $B$ . If the first radio

that you check is defective, what is the conditional probability that the other one is also defective?

**26.** Show that if  $P(A|B) = 1$ , then  $P(B^c|A^c) = 1$ .

**27.** An urn initially contains 1 red and 1 blue ball. At each stage, a ball is randomly withdrawn and replaced by two other balls of the same color. (For instance, if the red ball is initially chosen, then there would be 2 red and 1 blue balls in the urn when the next selection occurs.) Show by mathematical induction that the probability that there are exactly  $i$  red balls in the urn after  $n$  stages have been completed is  $\frac{1}{n+1}, 1 \leq i \leq n + 1$ .

**28.** A total of  $2n$  cards, of which 2 are aces, are to be randomly divided among two players, with each player receiving  $n$  cards. Each player is then to declare, in sequence, whether he or she has received any aces. What is the conditional probability that the second player has no aces, given that the first player declares in the affirmative, when (a)  $n = 2$ ? (b)  $n = 10$ ? (c)  $n = 100$ ? To what does the probability converge as  $n$  goes to infinity? Why?

**29.** There are  $n$  distinct types of coupons, and each coupon obtained is, independently of prior types collected, of type  $i$  with probability  $p_i, \sum_{i=1}^n p_i = 1$ .

(a) If  $n$  coupons are collected, what is the probability that one of each type is obtained?

(b) Now suppose that  $p_1 = p_2 = \dots = p_n = 1/n$ . Let  $E_i$  be the event that there are no type  $i$  coupons among the  $n$  collected. Apply the inclusion-exclusion identity for the probability of the union of events to  $P(\cup_i E_i)$  to prove the identity

$$n! = \sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)^n$$

**30.** Show that for any events  $E$  and  $F$ ,

$$P(E|E \cup F) \geq P(E|F)$$

*Hint:* Compute  $P(E|E \cup F)$  by conditioning on whether  $F$  occurs.

**31.** There is a 60 percent chance that event  $A$  will occur. If  $A$  does not occur, then there is a 10 percent chance that  $B$  will occur.

What is the probability that at least one of the events  $A$  or  $B$  will occur?

## Answers to Selected Problems

- |               |                                      |                            |                  |                     |                             |                                    |
|---------------|--------------------------------------|----------------------------|------------------|---------------------|-----------------------------|------------------------------------|
| <b>1.</b> 1/3 | <b>2.</b> 1/6; 1/5; 1/4; 1/3; 1/2; 1 | <b>3.</b> .339             | <b>5.</b> 6/91   | <b>16.</b> .9835    | <b>17.</b> .0792; .264      | <b>18.</b> .331; .383; .286; .4862 |
| <b>6.</b> 1/2 | <b>7.</b> 2/3                        | <b>8.</b> 1/2              | <b>9.</b> 7/11   | <b>10.</b> .22      | <b>11.</b> 1/17;            | <b>19.</b> 44.29; 41.18            |
| 1/33          | <b>12.</b> .504; .3629               | <b>14.</b> 35/768; 210/768 | <b>15.</b> .4848 | <b>20.</b> .4; 1/26 | <b>21.</b> .496; 3/14; 9/62 | <b>22.</b> 5/9; 1/6; 5/54          |
|               |                                      |                            |                  | <b>23.</b> 4/9; 1/2 | <b>24.</b> 1/3; 1/2         | <b>26.</b> 20/21;                  |

40/41    **28.** 3/128; 29/1536    **29.** .0893    **30.** 7/12; 3/5  
**33.** .76, 49/76    **34.** 27/31    **35.** .62, 10/19    **36.** 1/2  
**37.** 1/3; 1/5; 1    **38.** 12/37    **39.** 46/185    **40.** 3/13; 5/13;  
 5/52; 15/52    **41.** 43/459    **42.** 1.03 percent; .3046    **43.** 4/9  
**45.** 1/11    **48.** 2/3    **50.** .175; 38/165; 17/33    **51.** .65;  
 56/65; 8/65; 1/65; 14/35; 12/35; 9/35    **52.** .11; 16/89;  
 12/27; 3/5; 9/25    **55.** 9    **57.** (c) 2/3    **60.** 2/3; 1/3; 3/4  
**61.** 1/6; 3/20    **65.** .4375    **69.** 9; 9; 18; 110; 4; 4; 8; 120  
 all over 128    **70.** 1/9; 1/18    **71.** 38/64; 13/64; 13/64  
**73.** 1/16; 1/32; 5/16; 1/4; 31/32    **74.** 9/19    **75.** 3/4, 7/12  
**78.**  $2p^3(1-p) + 2p(1-p)^3; p^2/(1-2p+2p^2)$     **79.** .5550  
**81.** .9530    **83.** .5; .6; .8    **84.** 9/19; 6/19; 4/19; 7/15; 53/165;  
 7/33    **87.** 9/16    **90.** 97/142; 15/26; 33/102

## Solutions to Self-Test Problems and Exercises

$$1. (a) P(\text{no aces}) = \frac{\binom{35}{13}}{\binom{39}{13}}$$

$$(b) 1 - P(\text{no aces}) = \frac{4 \binom{35}{12}}{\binom{39}{13}}$$

$$(c) P(i \text{ aces}) = \frac{\binom{3}{i} \binom{36}{13-i}}{\binom{39}{13}}$$

**2.** Let  $L_i$  denote the event that the life of the battery is greater than  $10,000 \times i$  miles.

$$(a) P(L_2|L_1) = P(L_1L_2)/P(L_1) = P(L_2)/P(L_1) = 1/2$$

$$(b) P(L_3|L_1) = P(L_1L_3)/P(L_1) = P(L_3)/P(L_1) = 1/8$$

**3.** Put 1 white and 0 black balls in urn one, and the remaining 9 white and 10 black balls in urn two.

**4.** Let  $T$  be the event that the transferred ball is white, and let  $W$  be the event that a white ball is drawn from urn  $B$ . Then

$$\begin{aligned}
 P(T|W) &= \frac{P(W|T)P(T)}{P(W|T)P(T) + P(W|T^c)P(T^c)} \\
 &= \frac{(2/7)(2/3)}{(2/7)(2/3) + (1/7)(1/3)} = 4/5
 \end{aligned}$$

$$5. (a) P(E|E \cup F) = \frac{P(E \cap F)}{P(E \cup F)} = \frac{P(E)}{P(E) + P(F)}$$

since  $E \cap F = E$  and  $P(E \cup F) = P(E) + P(F)$  because  $E$  and  $F$  are mutually exclusive.

$$(b) P(E_j | \bigcup_{i=1}^{\infty} E_i) = \frac{P(E_j \cap \bigcup_{i=1}^{\infty} E_i)}{P(\bigcup_{i=1}^{\infty} E_i)} = \frac{P(E_j)}{\sum_{i=1}^{\infty} P(E_i)}$$

**6.** Let  $B_i$  denote the event that ball  $i$  is black, and let  $R_i = B_i^c$ . Then

$$\begin{aligned}
 P(B_1|R_2) &= \frac{P(R_2|B_1)P(B_1)}{P(R_2|B_1)P(B_1) + P(R_2|R_1)P(R_1)} \\
 &= \frac{[r/(b+r+c)][b/(b+r)]}{[r/(b+r+c)][b/(b+r)] + [(r+c)/(b+r+c)][r/(b+r)]} \\
 &= \frac{b}{b+r+c}
 \end{aligned}$$

**7.** Let  $B$  denote the event that both cards are aces.

$$\begin{aligned}
 (a) P(B|\text{yes to ace of spades}) &= \frac{P(B, \text{yes to ace of spades})}{P(\text{yes to ace of spades})} \\
 &= \frac{\frac{\binom{1}{1}\binom{3}{1}}{\binom{52}{2}}}{\frac{\binom{1}{1}\binom{51}{1}}{\binom{52}{2}}} \\
 &= 3/51
 \end{aligned}$$

**(b)** Since the second card is equally likely to be any of the remaining 51, of which 3 are aces, we see that the answer in this situation is also 3/51.

**(c)** Because we can always interchange which card is considered first and which is considered second, the result should be the same as in part (b). A more formal argument is as follows:

$$\begin{aligned}
 P(B|\text{second is ace}) &= \frac{P(B, \text{second is ace})}{P(\text{second is ace})} \\
 &= \frac{P(B)}{P(B) + P(\text{first is not ace, second is ace})} \\
 &= \frac{(4/52)(3/51)}{(4/52)(3/51) + (48/52)(4/51)} \\
 &= 3/51
 \end{aligned}$$

$$\begin{aligned}
 (d) P(B|\text{at least one}) &= \frac{P(B)}{P(\text{at least one})} \\
 &= \frac{(4/52)(3/51)}{1 - (48/52)(47/51)} \\
 &= 1/33
 \end{aligned}$$

$$8. \frac{P(H|E)}{P(G|E)} = \frac{P(HE)}{P(GE)} = \frac{P(H)P(E|H)}{P(G)P(E|G)}$$

Hypothesis  $H$  is 1.5 times as likely.

**9.** Let  $A$  denote the event that the plant is alive and let  $W$  be the event that it was watered.

$$\begin{aligned}
 (a) P(A) &= P(A|W)P(W) + P(A|W^c)P(W^c) \\
 &= (.85)(.9) + (.2)(.1) = .785
 \end{aligned}$$

$$\begin{aligned}
 (b) P(W^c|A^c) &= \frac{P(A^c|W^c)P(W^c)}{P(A^c)} \\
 &= \frac{(.8)(.1)}{.215} = \frac{16}{43}
 \end{aligned}$$

**10. (a)** Let  $R$  be the event that at least one red ball is chosen. Then

$$P(R) = 1 - P(R^c) = 1 - \frac{\binom{22}{6}}{\binom{30}{6}}$$

**(b)** Let  $G_2$  be the event there are exactly 2 green balls chosen. Working with the reduced sample space yields

$$P(G_2|R^c) = \frac{\binom{10}{2}\binom{12}{4}}{\binom{22}{6}}$$

**11.** Let  $W$  be the event that the battery works, and let  $C$  and  $D$  denote the events that the battery is a type  $C$  and that it is a type  $D$  battery, respectively.

**(a)**  $P(W) = P(W|C)P(C) + P(W|D)P(D) = .7(8/14) + .4(6/14) = 4/7$

**(b)**  $P(C|W^c) = \frac{P(W^c|C)P(C)}{P(W^c)} = \frac{.3(8/14)}{3/7} = .4$

**12.** Let  $L_i$  be the event that Maria likes book  $i, i = 1, 2$ . Then

$$P(L_2|L_1^c) = \frac{P(L_1^c L_2)}{P(L_1^c)} = \frac{P(L_1^c L_2)}{.4}$$

Using that  $L_2$  is the union of the mutually exclusive events  $L_1 L_2$  and  $L_1^c L_2$ , we see that

$$.5 = P(L_2) = P(L_1 L_2) + P(L_1^c L_2) = .4 + P(L_1^c L_2)$$

Thus,

$$P(L_2|L_1^c) = \frac{.1}{.4} = .25$$

**13. (a)** This is the probability that the last ball removed is blue. Because each of the 30 balls is equally likely to be the last one removed, the probability is  $1/3$ .

**(b)** This is the probability that the last red or blue ball to be removed is a blue ball. Because it is equally likely to be any of the 30 red or blue balls, the probability that it is blue is  $1/3$ .

**(c)** Let  $B_1, R_2, G_3$  denote, respectively, the events that the first color removed is blue, the second is red, and the third is green. Then

$$P(B_1 R_2 G_3) = P(G_3)P(R_2|G_3)P(B_1|R_2 G_3) = \frac{8}{38} \frac{20}{30} = \frac{8}{57}$$

where  $P(G_3)$  is just the probability that the very last ball is green and  $P(R_2|G_3)$  is computed by noting that given that the last ball is green, each of the 20 red and 10 blue balls is equally likely to be the last of that group to be removed, so the probability that it is one of the red balls is  $20/30$ . (Of course,  $P(B_1|R_2 G_3) = 1$ .)

**(d)**  $P(B_1) = P(B_1 G_2 R_3) + P(B_1 R_2 G_3) = \frac{20}{38} \frac{8}{18} + \frac{8}{57} = \frac{64}{171}$

**14.** Let  $H$  be the event that the coin lands heads, let  $T_h$  be the event that  $B$  is told that the coin landed heads, let  $F$  be the event that  $A$  forgets the result of the toss, and let  $C$  be the event that  $B$  is told the correct result. Then

$$\begin{aligned} \text{(a)} \quad P(T_h) &= P(T_h|F)P(F) + P(T_h|F^c)P(F^c) \\ &= (.5)(.4) + P(H)(.6) \\ &= .68 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad P(C) &= P(C|F)P(F) + P(C|F^c)P(F^c) \\ &= (.5)(.4) + 1(.6) = .80 \end{aligned}$$

$$\text{(c)} \quad P(H|T_h) = \frac{P(HT_h)}{P(T_h)}$$

Now,

$$\begin{aligned} P(HT_h) &= P(HT_h|F)P(F) + P(HT_h|F^c)P(F^c) \\ &= P(H|F)P(T_h|HF)P(F) + P(H)P(F^c) \\ &= (.8)(.5)(.4) + (.8)(.6) = .64 \end{aligned}$$

giving the result  $P(H|T_h) = .64/.68 = 16/17$ .

**15.** Since the black rat has a brown sibling, we can conclude that both of its parents have one black and one brown gene.

**(a)**  $P(2 \text{ black} | \text{at least one}) = \frac{P(2)}{P(\text{at least one})} = \frac{1/4}{3/4} = \frac{1}{3}$

**(b)** Let  $F$  be the event that all 5 offspring are black, let  $B_2$  be the event that the black rat has 2 black genes, and let  $B_1$  be the event that it has 1 black and 1 brown gene. Then

$$\begin{aligned} P(B_2|F) &= \frac{P(F|B_2)P(B_2)}{P(F|B_2)P(B_2) + P(F|B_1)P(B_1)} \\ &= \frac{(1)(1/3)}{(1)(1/3) + (1/2)^5(2/3)} = \frac{16}{17} \end{aligned}$$

**16.** Let  $F$  be the event that a current flows from  $A$  to  $B$ , and let  $C_i$  be the event that relay  $i$  closes. Then

$$P(F) = P(F|C_1)p_1 + P(F|C_1^c)(1 - p_1)$$

Now,

$$\begin{aligned} P(F|C_1) &= P(C_4 \cup C_2 C_5 \cup C_3 C_5) \\ &= p_4 + p_2 p_5 + p_3 p_5 - p_4 p_2 p_5 \\ &\quad - p_4 p_3 p_5 - p_2 p_3 p_5 + p_4 p_2 p_5 p_3 \end{aligned}$$

Also,

$$\begin{aligned} P(F|C_1^c) &= P(C_2 C_5 \cup C_2 C_3 C_4) \\ &= p_2 p_5 + p_2 p_3 p_4 - p_2 p_3 p_4 p_5 \end{aligned}$$

Hence, for part (a), we obtain

$$\begin{aligned} P(F) &= p_1(p_4 + p_2 p_5 + p_3 p_5 - p_4 p_2 p_5 \\ &\quad - p_4 p_3 p_5 - p_2 p_3 p_5 + p_4 p_2 p_5 p_3) \\ &\quad + (1 - p_1)p_2(p_5 + p_3 p_4 - p_3 p_4 p_5) \end{aligned}$$

For part (b), let  $q_i = 1 - p_i$ . Then

$$\begin{aligned} P(C_3|F) &= P(F|C_3)P(C_3)/P(F) \\ &= p_3[1 - P(C_1^c C_2^c \cup C_4^c C_5^c)]/P(F) \\ &= p_3(1 - q_1 q_2 - q_4 q_5 + q_1 q_2 q_4 q_5)/P(F) \end{aligned}$$

**17.** Let  $A$  be the event that component 1 is working, and let  $F$  be the event that the system functions.

**(a)**  $P(A|F) = \frac{P(AF)}{P(F)} = \frac{P(A)}{1 - (1/2)^2} = \frac{2}{3}$

where  $P(F)$  was computed by noting that it is equal to 1 minus the probability that components 1 and 2 are both failed.

**(b)**  $P(A|F) = \frac{P(AF)}{P(F)} = \frac{P(F|A)P(A)}{P(F)} = \frac{(3/4)(1/2)}{(1/2)^3 + 3(1/2)^3} = \frac{3}{4}$

where  $P(F)$  was computed by noting that it is equal to the probability that all 3 components work plus the three probabilities relating to exactly 2 of the components working.

**18.** If we assume that the outcomes of the successive spins are independent, then the conditional probability of the next outcome is unchanged by the result that the previous 10 spins landed on black.

**19.** Condition on the outcome of the initial tosses:

$$\begin{aligned} P(A \text{ odd}) &= P_1(1 - P_2)(1 - P_3) + (1 - P_1)P_2P_3 \\ &\quad + P_1P_2P_3P(A \text{ odd}) \\ &\quad + (1 - P_1)(1 - P_2)(1 - P_3)P(A \text{ odd}) \end{aligned}$$

so,

$$P(A \text{ odd}) = \frac{P_1(1 - P_2)(1 - P_3) + (1 - P_1)P_2P_3}{P_1 + P_2 + P_3 - P_1P_2 - P_1P_3 - P_2P_3}$$

**20.** Let  $A$  and  $B$  be the events that the first trial is larger and that the second is larger, respectively. Also, let  $E$  be the event that the results of the trials are equal. Then

$$1 = P(A) + P(B) + P(E)$$

But, by symmetry,  $P(A) = P(B)$ : thus,

$$P(B) = \frac{1 - P(E)}{2} = \frac{1 - \sum_{i=1}^n p_i^2}{2}$$

Another way of solving the problem is to note that

$$\begin{aligned} P(B) &= \sum_i \sum_{j>i} P(\text{first trial results in } i, \text{ second trial results in } j) \\ &= \sum_i \sum_{j>i} p_i p_j \end{aligned}$$

To see that the two expressions derived for  $P(B)$  are equal, observe that

$$\begin{aligned} 1 &= \sum_{i=1}^n p_i \sum_{j=1}^n p_j \\ &= \sum_i \sum_j p_i p_j \\ &= \sum_i p_i^2 + \sum_i \sum_{j \neq i} p_i p_j \\ &= \sum_i p_i^2 + 2 \sum_i \sum_{j>i} p_i p_j \end{aligned}$$

**21.** Let  $E = \{A \text{ gets more heads than } B\}$ ; then

$$\begin{aligned} P(E) &= P(E|A \text{ leads after both flip } n)P(A \text{ leads after both flip } n) \\ &\quad + P(E|\text{even after both flip } n)P(\text{even after both flip } n) \\ &\quad + P(E|B \text{ leads after both flip } n)P(B \text{ leads after both flip } n) \\ &= P(A \text{ leads}) + \frac{1}{2}P(\text{even}) \end{aligned}$$

Now, by symmetry,

$$\begin{aligned} P(A \text{ leads}) &= P(B \text{ leads}) \\ &= \frac{1 - P(\text{even})}{2} \end{aligned}$$

Hence,

$$P(E) = \frac{1}{2}$$

**22. (a)** Not true: In rolling 2 dice, let  $E = \{\text{sum is } 7\}$ ,  $F = \{\text{1st die does not land on } 4\}$ , and  $G = \{\text{2nd die does not land on } 3\}$ . Then

$$P(E|F \cup G) = \frac{P(\{7, \text{not } (4, 3)\})}{P(\text{not } (4, 3))} = \frac{5/36}{35/36} = 5/35 \neq P(E)$$

**(b)**  $P(E(F \cup G)) = P(EF \cup EG)$   
 $= P(EF) + P(EG)$  since  $EFG = \emptyset$   
 $= P(E)[P(F) + P(G)]$   
 $= P(E)P(F \cup G)$  since  $FG = \emptyset$

**(c)**  $P(G|EF) = \frac{P(EFG)}{P(EF)}$   
 $= \frac{P(E)P(FG)}{P(EF)}$  since  $E$  is independent of  $FG$   
 $= \frac{P(E)P(F)P(G)}{P(E)P(F)}$  by independence  
 $= P(G).$

**23. (a)** necessarily false; if they were mutually exclusive, then we would have

$$0 = P(AB) \neq P(A)P(B)$$

**(b)** necessarily false; if they were independent, then we would have

$$P(AB) = P(A)P(B) > 0$$

(c) necessarily false; if they were mutually exclusive, then we would have

$$P(A \cup B) = P(A) + P(B) = 1.2$$

(d) possibly true

**24.** The probabilities in parts (a), (b), and (c) are .5, (.8)<sup>3</sup> = .512, and (.9)<sup>7</sup> ≈ .4783, respectively.

**25.** Let  $D_i, i = 1, 2$ , denote the event that radio  $i$  is defective. Also, let  $A$  and  $B$  be the events that the radios were produced at factory  $A$  and at factory  $B$ , respectively. Then

$$\begin{aligned} P(D_2|D_1) &= \frac{P(D_1 D_2)}{P(D_1)} \\ &= \frac{P(D_1 D_2|A)P(A) + P(D_1 D_2|B)P(B)}{P(D_1|A)P(A) + P(D_1|B)P(B)} \\ &= \frac{(.05)^2(1/2) + (.01)^2(1/2)}{(.05)(1/2) + (.01)(1/2)} \\ &= 13/300 \end{aligned}$$

**26.** We are given that  $P(AB) = P(B)$  and must show that this implies that  $P(B^c A^c) = P(A^c)$ . One way is as follows:

$$\begin{aligned} P(B^c A^c) &= P((A \cup B)^c) \\ &= 1 - P(A \cup B) \\ &= 1 - P(A) - P(B) + P(AB) \\ &= 1 - P(A) \\ &= P(A^c) \end{aligned}$$

**27.** The result is true for  $n = 0$ . With  $A_i$  denoting the event that there are  $i$  red balls in the urn after stage  $n$ , assume that

$$P(A_i) = \frac{1}{n+1}, \quad i = 1, \dots, n+1$$

Now let  $B_j, j = 1, \dots, n+2$ , denote the event that there are  $j$  red balls in the urn after stage  $n+1$ . Then

$$\begin{aligned} P(B_j) &= \sum_{i=1}^{n+1} P(B_j|A_i)P(A_i) \\ &= \frac{1}{n+1} \sum_{i=1}^{n+1} P(B_j|A_i) \\ &= \frac{1}{n+1} [P(B_j|A_{j-1}) + P(B_j|A_j)] \end{aligned}$$

Because there are  $n+2$  balls in the urn after stage  $n$ , it follows that  $P(B_j|A_{j-1})$  is the probability that a red ball is chosen when  $j-1$  of the  $n+2$  balls in the urn are red and  $P(B_j|A_j)$  is the probability that a red ball is not chosen when  $j$  of the  $n+2$  balls in the urn are red. Consequently,

$$P(B_j|A_{j-1}) = \frac{j-1}{n+2}, \quad P(B_j|A_j) = \frac{n+2-j}{n+2}$$

Substituting these results into the equation for  $P(B_j)$  gives

$$P(B_j) = \frac{1}{n+1} \left[ \frac{j-1}{n+2} + \frac{n+2-j}{n+2} \right] = \frac{1}{n+2}$$

This completes the induction proof.

**28.** If  $A_i$  is the event that player  $i$  receives an ace, then

$$P(A_i) = 1 - \frac{\binom{2n-2}{n}}{\binom{2n}{n}} = 1 - \frac{1}{2} \frac{n-1}{2n-1} = \frac{3n-1}{4n-2}$$

By arbitrarily numbering the aces and noting that the player who does not receive ace number one will receive  $n$  of the remaining  $2n-1$  cards, we see that

$$P(A_1 A_2) = \frac{n}{2n-1}$$

Therefore,

$$P(A_2^c|A_1) = 1 - P(A_2|A_1) = 1 - \frac{P(A_1 A_2)}{P(A_1)} = \frac{n-1}{3n-1}$$

We may regard the card division outcome as the result of two trials, where trial  $i, i = 1, 2$ , is said to be a success if ace number  $i$  goes to the first player. Because the locations of the two aces become independent as  $n$  goes to infinity, with each one being equally likely to be given to either player, it follows that the trials become independent, each being a success with probability  $1/2$ . Hence, in the limiting case where  $n \rightarrow \infty$ , the problem becomes one of determining the conditional probability that two heads result, given that at least one does, when two fair coins are flipped. Because  $\frac{n-1}{3n-1}$  converges to  $1/3$ , the answer agrees with that of Example 2b.

**29. (a)** For any permutation  $i_1, \dots, i_n$  of  $1, 2, \dots, n$ , the probability that the successive types collected is  $i_1, \dots, i_n$  is  $p_{i_1} \cdots p_{i_n} = \prod_{i=1}^n p_i$ . Consequently, the desired probability is  $n! \prod_{i=1}^n p_i$ .

**(b)** For  $i_1, \dots, i_k$  all distinct,

$$P(E_{i_1} \cdots E_{i_k}) = \left( \frac{n-k}{n} \right)^n$$

which follows because there are no coupons of types  $i_1, \dots, i_k$  when each of the  $n$  independent selections is one of the other  $n-k$  types. It now follows by the inclusion-exclusion identity that

$$P(\cup_{i=1}^n E_i) = \sum_{k=1}^n (-1)^{k+1} \binom{n}{k} \left( \frac{n-k}{n} \right)^n$$

Because  $1 - P(\cup_{i=1}^n E_i)$  is the probability that one of each type is obtained, by part (a) it is equal to  $\frac{n!}{n^n}$ . Substituting this into the preceding equation gives

$$1 - \frac{n!}{n^n} = \sum_{k=1}^n (-1)^{k+1} \binom{n}{k} \left( \frac{n-k}{n} \right)^n$$

or

$$n! = n^n - \sum_{k=1}^n (-1)^{k+1} \binom{n}{k} (n-k)^n$$

or

$$n! = \sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)^n$$

$$\begin{aligned} \mathbf{30.} \quad P(E|E \cup F) &= P(E|F(E \cup F))P(F|E \cup F) \\ &\quad + P(E|F^c(E \cup F))P(F^c|E \cup F) \end{aligned}$$

Using

$$F(E \cup F) = F \quad \text{and} \quad F^c(E \cup F) = F^c E$$

gives

$$\begin{aligned} P(E|E \cup F) &= P(E|F)P(F|E \cup F) + P(E|F^c E)P(F^c|E \cup F) \\ &= P(E|F)P(F|E \cup F) + P(F^c|E \cup F) \\ &\geq P(E|F)P(F|E \cup F) + P(E|F)P(F^c|E \cup F) \\ &= P(E|F) \end{aligned}$$

$$\begin{aligned} \mathbf{31.} \quad P(A \cup B) &= P(A \cup B|A)P(A) + P(A \cup B|A^c)P(A^c) \\ &= 1(.6) + .1(.4) = .64 \end{aligned}$$